M.TECH · CS-6314

Information Architecture — Visual Study Book

UX/UI → Information Architecture → Organizing → Labeling & Search → TOGAF.
Every topic explained with a diagram, a real-life use case, an exam tip and a Hinglish memory hook.

Unit I · UX / UI
Unit II · Introduction to IA
Unit III · Organizing Information
Unit IV · Labeling & Search
Unit V · TOGAF

63 original diagrams · definitions to open every answer · Hinglish memory hooks · revision checklist

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How to use this book

The 3-step exam answer formula

⭐ **Exam tip — Write every answer in this order**

1. **Definition** — one line (the blue 📘 boxes give you this ready-made).
2. **Explain in 3–5 lines** — plain English, use the keywords in **bold**.
3. **Diagram or example** — draw the figure from the book; for comparison questions finish with a table.

Box guide

Box	What it gives you
📘 Definition	Opening line for the answer
🏭 Real-life use case	Where this is used in real projects — good for "applications" and viva questions
💡 Why it matters	The reason a concept exists (examiners love "why")
⭐ Exam tip	Marks-scoring points, what to memorise exactly
🧠 Yaad rakho	Short Hinglish memory hook
⚠️ Careful	Common mistakes and traps
❓ Likely exam questions	Past-paper style questions for that unit

Everything comes from your CS-6314 notes. Where this book adds an example or explanation that is **not** in your notes, it says "illustration" or "not in your notes".

Course outcomes

CO	Description (from your notes)
CO1	Develop conceptual understanding of UX/UI design and information architecture.
CO2	Build a theoretical foundation on principles and concepts of designing effective information architecture.
CO3	Study existing portals on organizing, classification, navigation, searching & labeling; prepare comprehensive strength/weakness reports.
CO4	Understand TOGAF for building information architecture documents.
CO5	Supplement fundamentals with case studies of modern web infrastructure — Cloud, Security and Mobile.

The notes do not map each CO to a unit, so this book does not either. CO5 (cloud, security, mobile case studies) has no topic section in the notes.

Syllabus map

Unit	Focus	Main figures to practise
I	UX vs UI, design issues, UX/UI process and tools, design principles	UX process · design thinking · 8 principles
II	What IA is, 8 IA principles, the information architect, areas, architecture types, research & card sorting	8 principles · four systems · architecture types · card sorting
III	Organization schemes, websites vs intranets, cohesive systems, WWW & browser, context, navigation	Exact vs ambiguous · 4 navigation types · integrated vs remote
IV	Labeling systems, creating and tuning labels, search behaviour, search interface, indexing, to search or not	Label types · search behaviours · search UI
V	TOGAF 9: components, ADM and its phases, deliverables, Enterprise Continuum, governance, views, building blocks	ADM cycle · continuum · view vs viewpoint

The big picture — how all five units fit together

Unit I is the whole experience; Unit II shows IA as its structural foundation; Units III and IV build the four systems of IA; Unit V zooms out to enterprise-level architecture.

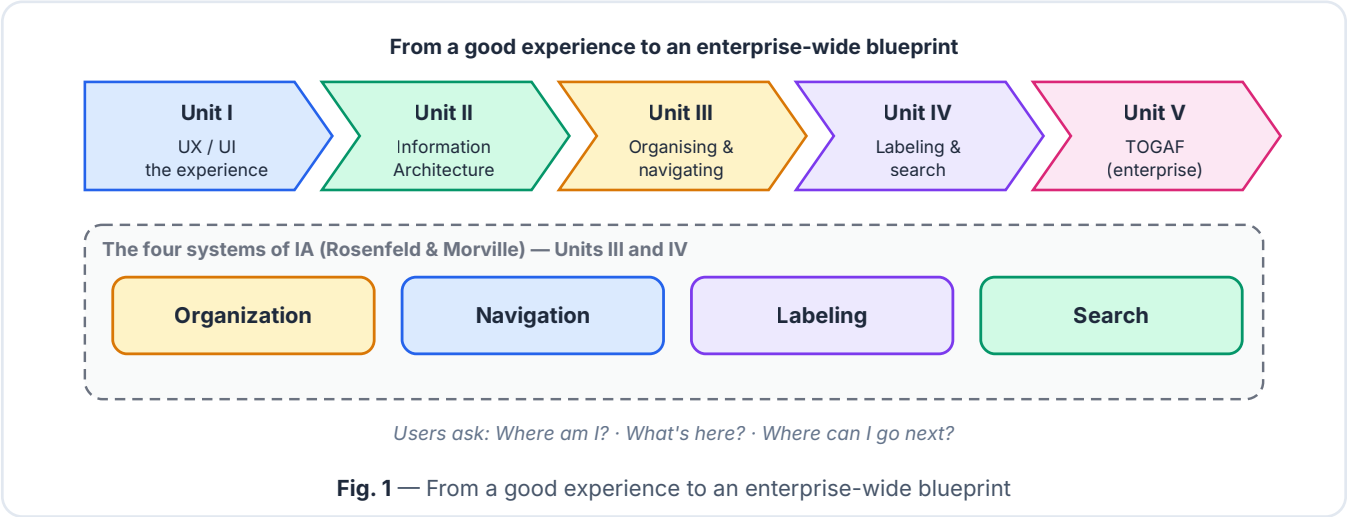


Fig. 1 — From a good experience to an enterprise-wide blueprint

🏭 Real-life use case — One portal, end to end

A university portal is redesigned. UX research finds students cannot find the syllabus (Unit I). The information architect restructures the content and defines the sitemap (Unit II), chooses organization and navigation systems (Unit III), rewrites labels and adds a search box (Unit IV). At the enterprise level, the IT office uses TOGAF to align such portals with university strategy (Unit V).

Practice reading: GeeksforGeeks UI/UX hub — geeksforgeeks.org (UI/UX) · The Open Group TOGAF — opengroup.org/togaf · Rosenfeld & Morville, *Information Architecture for the World Wide Web* (O'Reilly)

UNIT I

Introduction to UX/UI

UI vs UX | Design issues | UX/UI process | Tools | Design Thinking | Design principles | Visual design

1.1 What is UX and UI?

📘 **Definition**

User Interface (UI) is everything a user visually interacts with on a screen — buttons, icons, typography, spacing, colour and layout; it concerns how a product **looks**.
User Experience (UX) is the overall feeling and journey a person has while using a product — how easy and satisfying it is, and whether it solves their problem; it concerns how a product **works and feels**.

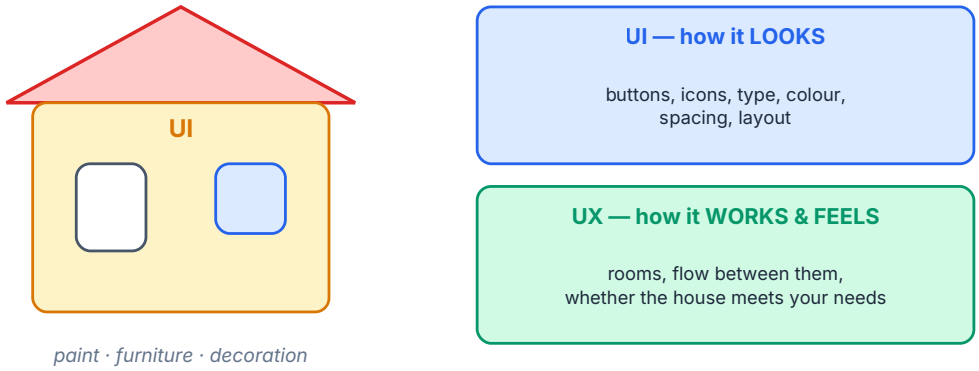


Fig. 2 — The house analogy — UI is the paint and furniture, UX is the layout and how well it suits you

Analogy: if a website is a house, UI is the paint, furniture and decoration, while UX is the layout of rooms, how easily you move between them and whether the house meets your needs.

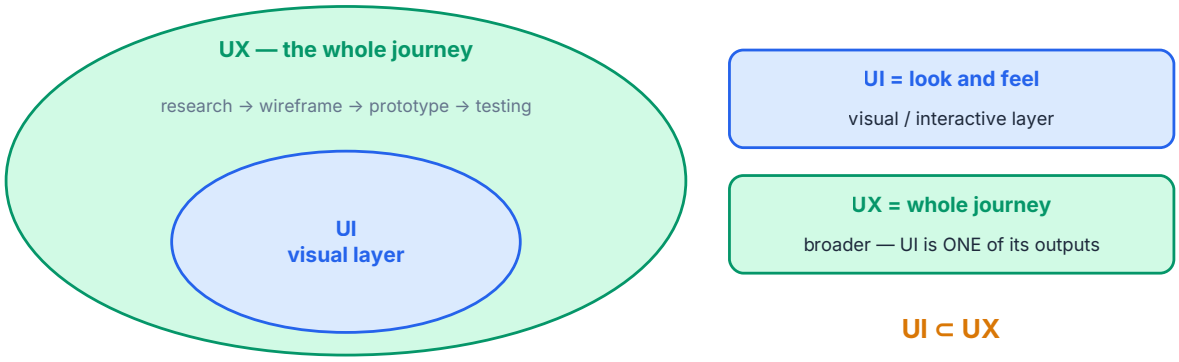


Fig. 3 — UI is the visual layer inside the broader UX journey

Point	UI	UX
Concerned with	How it looks	How it works and feels
Scope	Visual / interactive layer (look and feel)	The entire journey: research → wireframe → prototype → testing
Relationship	One output of UX	Broader — includes UI

🏭 Real-life use case — Same app, two views

A food-delivery app with beautiful buttons (good UI) can still have poor UX if reordering your last meal takes seven taps. The reverse also happens: a plain page can have great UX if it gets you to your task fast.

🧠 Yaad rakho

UI = **kaisa dikhta hai**. UX = **kaisa lagta hai / kaam karta hai**. UX bada, UI uska ek hissa.

📖 GFG: Why to use UI/UX Design? · UX Design — key process, flow and principles

1.2 Design Issues in UX/UI

📘 Definition

Design issues are the common problems a designer must anticipate and resolve while designing UX/UI.

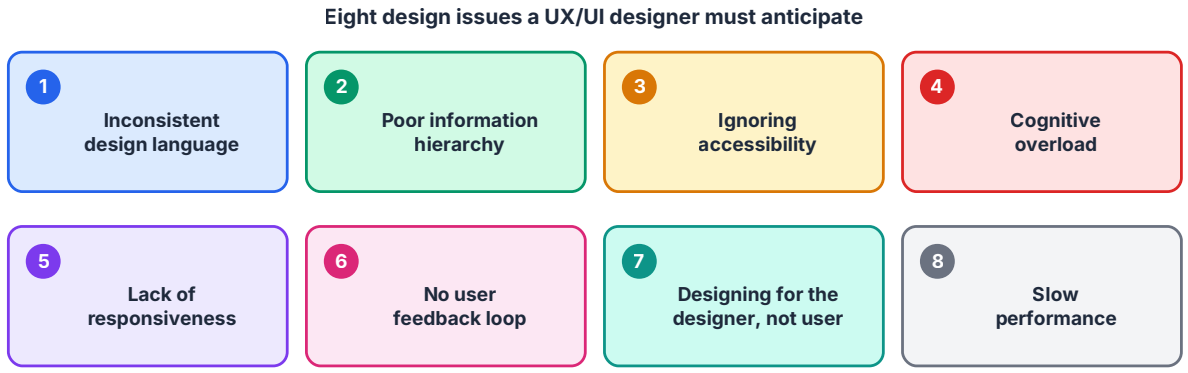


Fig. 4 — Eight common design issues

#	Issue	What goes wrong
1	**Inconsistent design language**	Different fonts, colours and button styles in one product confuse users
2	**Poor information hierarchy**	Everything has equal visual weight, so users can't tell what's important
3	**Ignoring accessibility**	Excludes users with visual, auditory, motor or cognitive disabilities
4	**Cognitive overload**	Too many options or information on one screen
5	**Lack of responsiveness**	Design breaks across mobile, tablet and desktop
6	**Ignoring feedback loops**	No confirmation after an action, e.g. form submission
7	**Designing for the designer**	Personal taste instead of user research and testing
8	**Slow performance**	Even a beautiful UI fails if load times are high



Fig. 5 — Poor hierarchy gives every element equal weight; good hierarchy uses size, colour and position

⭐ **Exam tip**

These issues are addressed through **usability testing, heuristic evaluation and established design principles** (Sec. 1.4). Write all three remedies at the end of your answer.

🧠 **Yaad rakho**

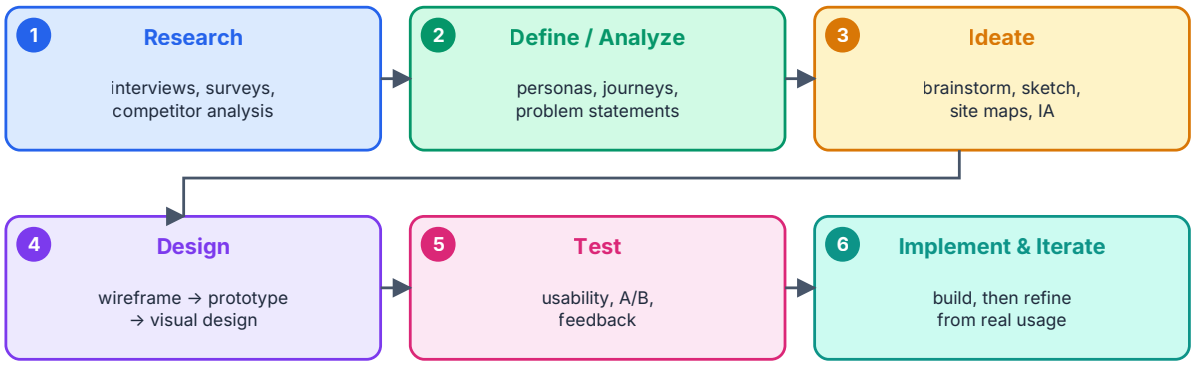
8 issues: "**Consistency, Hierarchy, Accessibility, Overload, Responsive, Feedback, Designer-bias, Speed**".

📖 GFG: Fundamentals of solid UI/UX design · Designing user-friendly interfaces

1.3 Tools and Processes of UX/UI Design

📘 **Definition**

The **UX/UI design process** is the sequence of stages — research, define, ideate, design, test, implement & iterate — through which a product moves from understanding users to a refined release.



Tools — Research/wireframe: Figma, Adobe XD, Sketch, Balsamiq · Prototype: InVision, Framer
Testing: Maze, UsabilityHub, Hotjar · Handoff: Zeplin, Figma Dev Mode, Miro

Fig. 6 — The six stages of the UX/UI design process, with common tools

1. **Research** — interviews, surveys, competitor analysis.
2. **Define / Analyze** — personas, user journeys, problem statements.
3. **Ideate** — brainstorming, sketching, information architecture and site maps.
4. **Design** — low-fidelity wireframe → interactive prototype → visual design.
5. **Test** — usability testing, A/B testing, feedback.
6. **Implement & Iterate** — developers build; designers keep refining from real usage data.

Stage / purpose	Tools in the notes
Research & wireframing	Figma, Adobe XD, Sketch, Balsamiq
Prototyping	Figma, InVision, Adobe XD, Framer
User testing	Maze, UsabilityHub, Hotjar
Collaboration / handoff	Zeplin, Figma Dev Mode, Miro (journey mapping)

Design Thinking

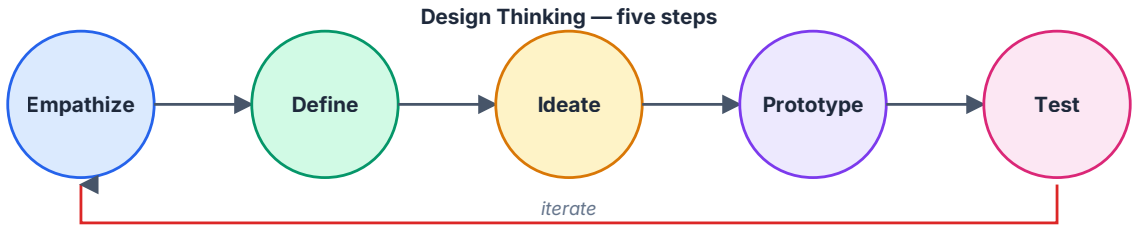


Fig. 7 — Design Thinking — five steps that map onto the UX process

The process maps directly to **Design Thinking**: Empathize → Define → Ideate → Prototype → Test.

🏭 Real-life use case — Redesigning a bus-booking app

The team **interviews** daily commuters (research), turns findings into a "Ravi the office-goer" **persona** (define), **sketches** a one-tap rebooking flow (ideate), builds a clickable **Figma prototype** (design), runs it with 5 users in **Maze** (test) and ships, then watches usage data (iterate). *(illustration, not from the notes)*

🧠 **Yaad rakho**

UX process: "**R-D-I-D-T-I**". Design Thinking: "**Empathize, Define, Ideate, Prototype, Test**" — pehle logon ko samjho, phir banao.

📖 GFG: Design Thinking principles in UI/UX · UX design key process

1.4 Design Principles of Good UX/UI Design

📘 **Definition**

Design principles are the core guidelines — simplicity, user-centred design, visibility, consistency, feedback, clarity, accessibility, efficiency and hierarchy — that make an interface usable.

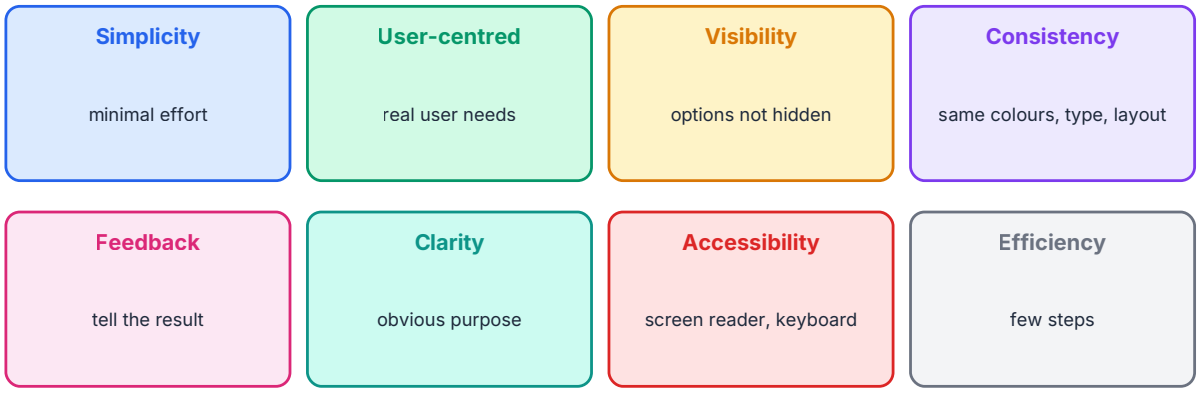
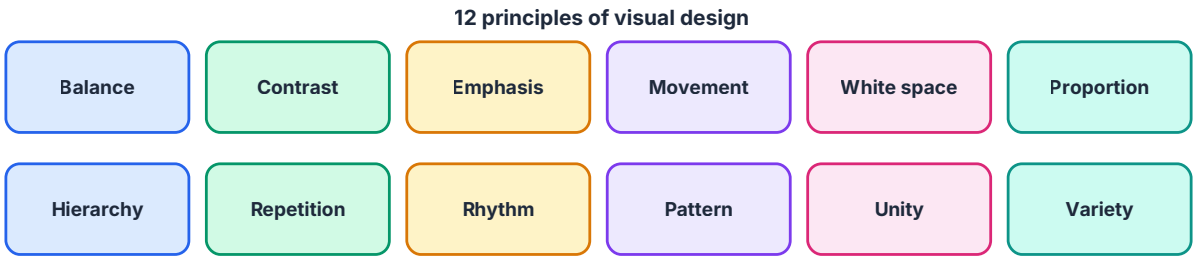


Fig. 8 — Eight design principles, plus hierarchy

Principle	Meaning
Simplicity	Minimal effort to use and understand
User-centred design	Designed around actual user needs, including users with disabilities
Visibility	Important tasks and options are clearly visible, not hidden
Consistency	Same colours, typography and layout patterns throughout — builds trust, reduces learning curve
Feedback	The system tells the user the result of their action (e.g. success message)
Clarity	Users immediately understand what a screen/element does
Accessibility	Usable by everyone, including screen-reader and keyboard-only users
Efficiency	Tasks completed quickly, with minimal steps
Hierarchy	Visual weight (size, colour, position) shows what matters most

12 principles of visual design



Handy for a diagram-based answer

Fig. 9 — The 12 principles of visual design

Commonly asked and useful for a diagram-based answer: balance, contrast, emphasis, movement, white space, proportion, hierarchy, repetition, rhythm, pattern, unity, variety.

> ⭐ **Exam tip**
>
> For "explain design principles with examples", pick 5–6 principles and give one concrete example each (e.g. Feedback → "Your form was submitted" message; Consistency → same button style on every page).

> ⚠️ **Careful**
>
> Do not mix up **Visibility** (options clearly shown) with **Feedback** (system responds to an action). One is before the action, the other after.

> ❓ **Likely exam questions**
>
> - Differentiate UX and UI with a suitable analogy.
> - Discuss the design issues a UX/UI designer must handle.
> - Explain the UX/UI design process with tools used at each stage.
> - Explain Design Thinking and relate it to the UX process.
> - Explain the design principles of good UX/UI with examples.
> - List the 12 principles of visual design.

📖 GFG: Principles of UI/UX design · 12 principles of visual design

UNIT II

Introduction to Information Architecture

What is IA | 8 principles | Information architect | Areas of IA | System / Enterprise / Application / Internet | Research & practice | Card sorting

2.1 What is Information Architecture?

Definition

Information Architecture (IA) is the practice of **organizing, structuring and labeling content** so that people can find information and complete tasks efficiently.

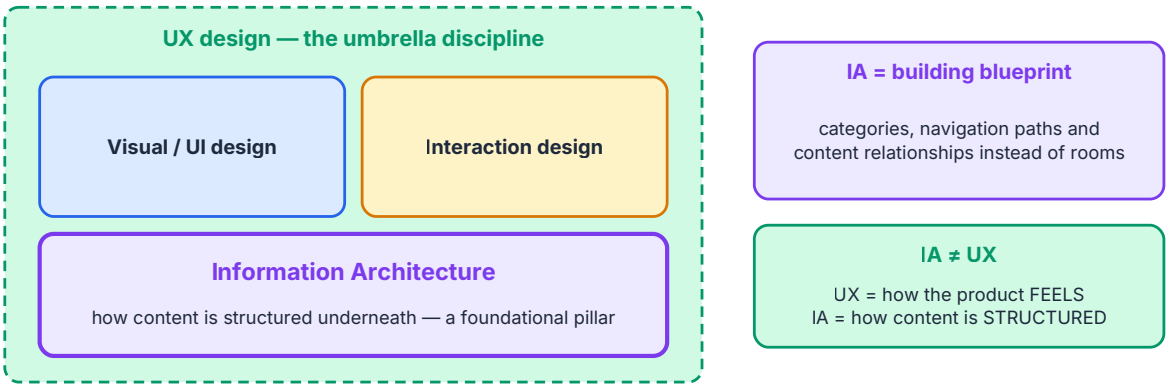


Fig. 10 — IA is a foundational pillar inside the wider UX discipline

Think of IA as the **blueprint of a building** — but instead of rooms and hallways it defines categories, navigation paths and content relationships within a website, app or system.

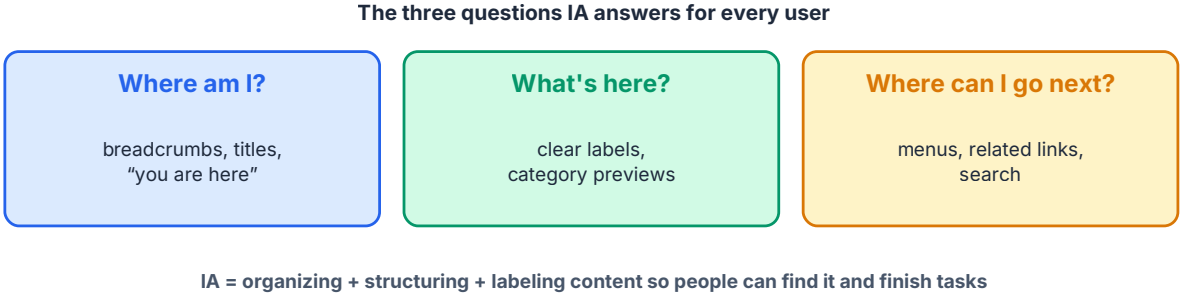


Fig. 11 — The three core questions IA answers for the user

IA answers three questions: **Where am I? What's here? Where can I go next?**

💡 **Why it matters**

IA is closely tied to but distinct from UX: **UX** is the umbrella discipline (how the product feels overall); **IA** is one of its foundational pillars (how content is structured underneath).

🏥 **Real-life use case — A hospital website**

Patients visit to book a doctor, find visiting hours or download reports. IA decides that "Departments", "Book Appointment" and "Reports" exist, how they group and where they link — before any colour or font is chosen.

🧠 **Yaad rakho**

IA = **content ka naksha**. Teen sawaal: "Main kahan hu? Yahan kya hai? Aage kahan jaun?"

📖 GFG: IA — a complete guide · IA in UX design · IA vs UX design

2.2 Principles of Information Architecture

📘 **Definition**

The **8 principles of IA** (Dan Brown's IA principles, foundational for this subject) are guidelines for structuring content so it stays findable, understandable and scalable.

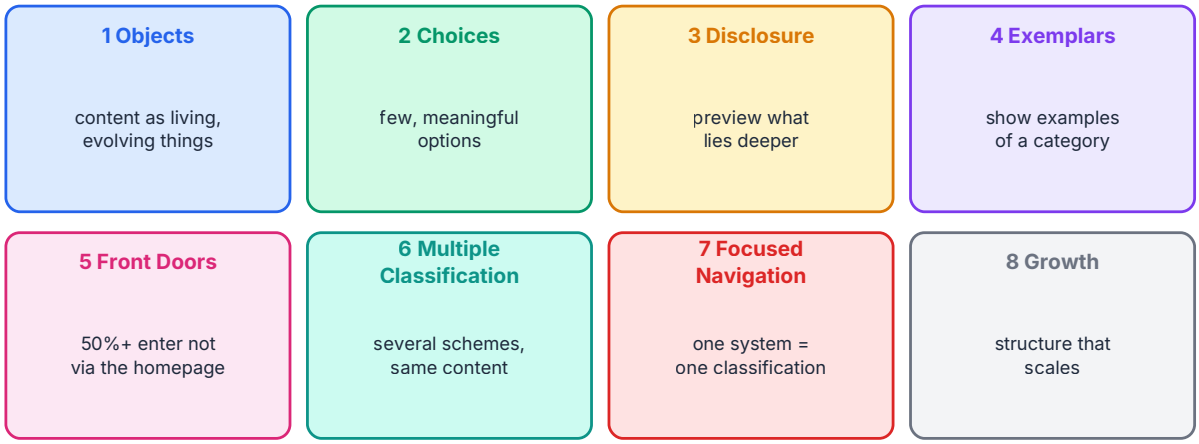


Fig. 12 — The eight IA principles

#	Principle	Meaning
1	**Objects**	Treat content as living, evolving objects with their own lifecycle, attributes and behaviours
2	**Choices**	Keep menus and choices small and meaningful; too many options overwhelm
3	**Disclosure**	Show enough information to help users understand what lies deeper (progressive disclosure)
4	**Exemplars**	Describe the contents of categories by giving examples of the content
5	**Front Doors**	Assume at least 50% of users enter through a page other than the homepage — every page must give context
6	**Multiple Classification**	Offer several classification schemes so users with different mental models can find the same content
7	**Focused Navigation**	Keep each navigation system focused on one type of classification, not mixed schemes
8	**Growth**	Design the structure so it scales as more content is added

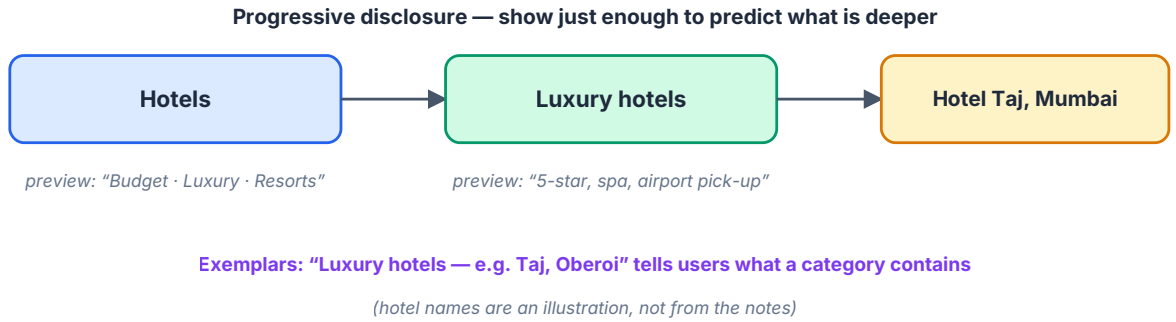


Fig. 13 — Disclosure and exemplars working together

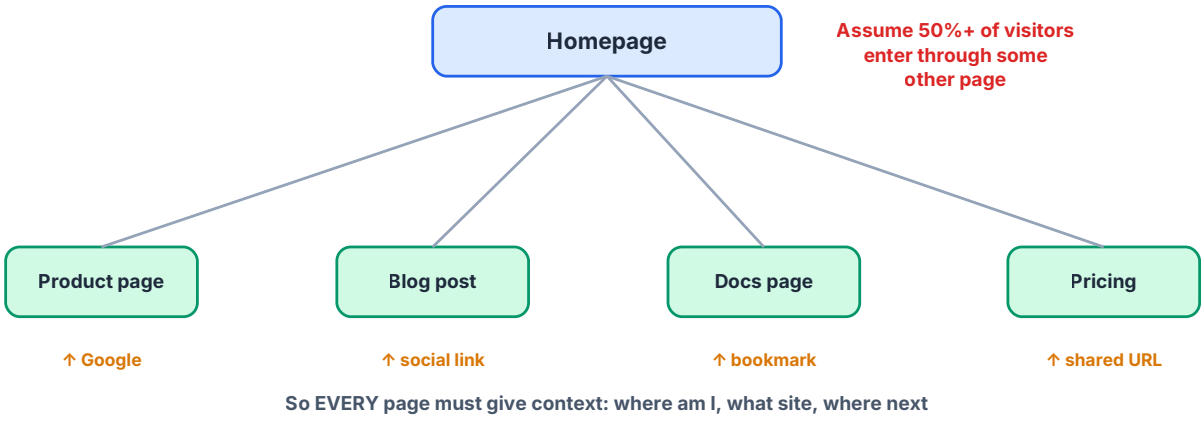


Fig. 14 — Front Doors — visitors often arrive on any page, not the homepage

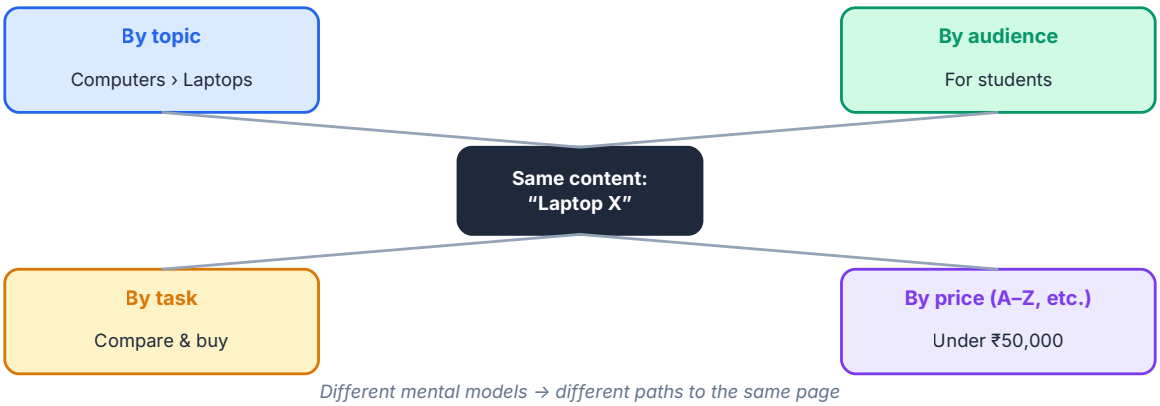


Fig. 15 — Multiple Classification — different schemes lead to the same content

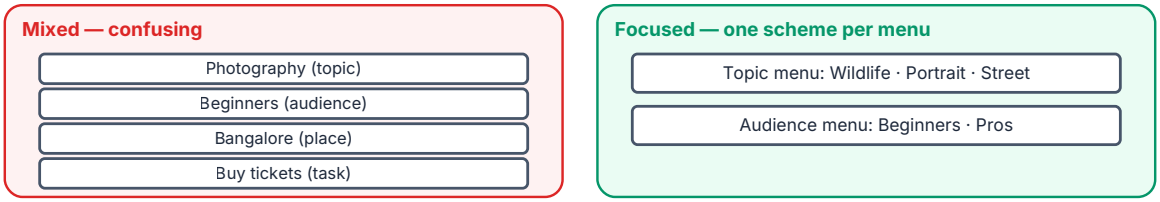


Fig. 16 — Focused Navigation — keep each system on one scheme

> ⚠️ **Careful**
>
> The examples in these four figures (hotels, laptops, photography menus) are illustrations to make the principles concrete — the notes give only the definitions.

> ⭐ **Exam tip**
>
> Write the principles in the standard order with one-line meanings; that alone usually earns most of the marks. Highlight the number **50%** for Front Doors.

> 🧠 **Yaad rakho**
>
> 8 principles: "**O-C-D-E-F-M-F-G**" — Objects, Choices, Disclosure, Exemplars, Front doors, Multiple classification, Focused navigation, Growth.

📖 GFG: IA — a complete guide · Dan Brown's 8 principles — *Information Architecture: a practical guide*

2.3 Role of the Information Architect

> 📘 **Definition**
>
> An **Information Architect** is the person responsible for organizing content so it makes sense to users and supports business goals.

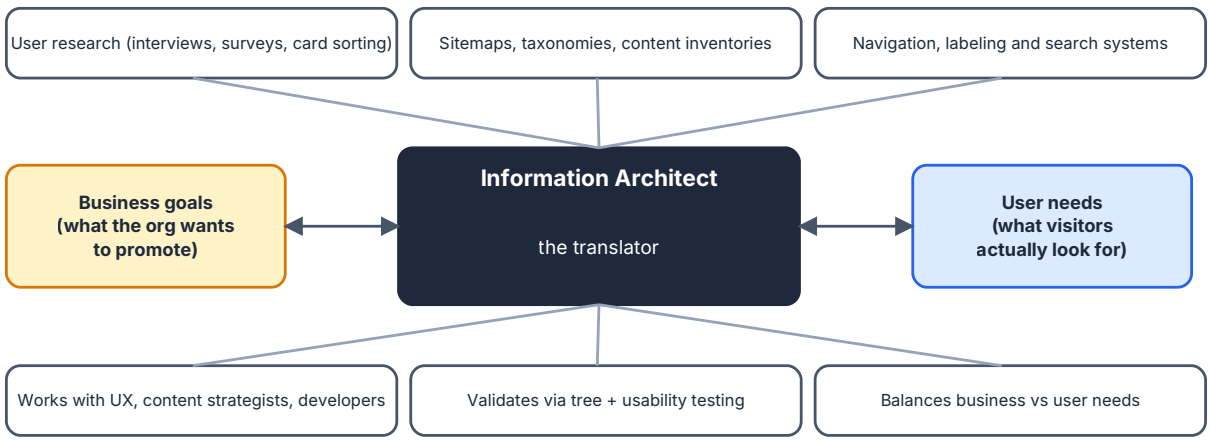


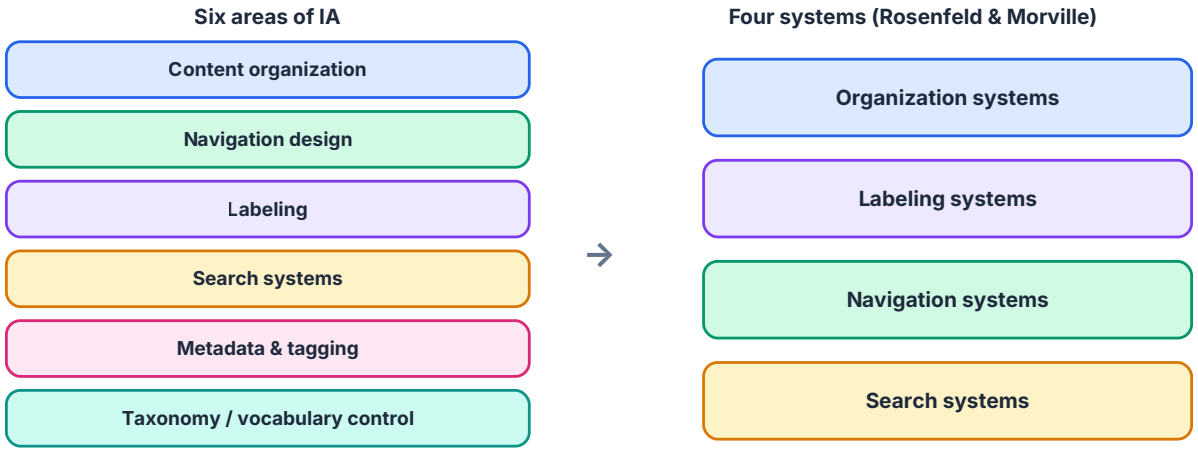
Fig. 17 — The information architect as translator between business goals and user needs

- Conducts **user research** (interviews, surveys, card sorting) to understand mental models.
- Creates **sitemaps, taxonomies and content inventories**.
- Defines **navigation, labeling and search** systems.
- Works closely with UX designers, content strategists, developers and stakeholders.
- Validates structures through **tree testing and usability testing**.
- Balances **business requirements** (what the organisation wants to promote) against **user needs** (what visitors look for).

> 🧠 **Yaad rakho**
>
> IA = **anuvadak (translator)** — business ki bhasha aur user ki bhasha ke beech.

📖 GFG: IA — a complete guide · Loop11 — A beginner's guide to IA in UX

2.4 Areas of Information Architecture



The 4-way split recurs throughout Units III and IV

Fig. 18 — Six areas of IA and the four systems they map onto

1. **Content organization** — grouping and categorising content logically.
2. **Navigation design** — menus, links, breadcrumbs for moving around.
3. **Labeling** — the words/icons representing categories and links.

4. **Search systems** — letting users find content via queries.
5. **Metadata & tagging** — descriptive data attached to content for filtering/search.
6. **Taxonomy / vocabulary control** — consistent terms across the site.

> 💡 **Why it matters — The four systems**
>
> These areas map onto the **four systems of IA** defined by Rosenfeld & Morville, the standard IA textbook authors: **Organization, Labeling, Navigation and Search systems**. This 4-way split recurs throughout Units III and IV.

📖 GFG: IA in UX design · Big Sea — What is Information Architecture?

2.5 Types of Architecture: System, Enterprise, Application, Internet

> 📘 **Definition**
>
> These four architecture types differ in **scope**: System = one system; Application = one application's internals; Enterprise = the whole organisation; Internet = networked, distributed systems on the web. **Frequently confused, high-weightage.**

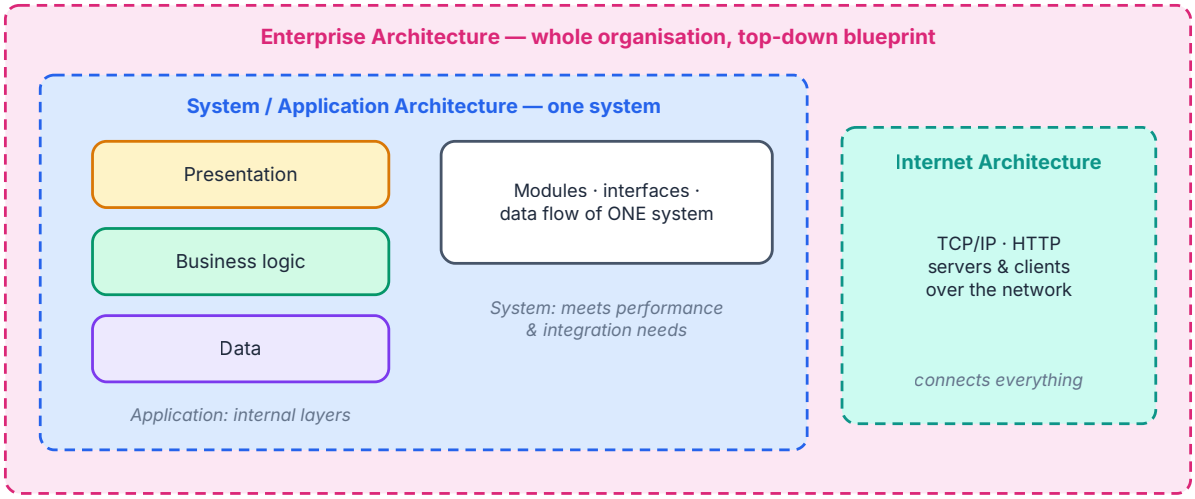


Fig. 19 — Scope of each architecture type

Type	Scope	Focus
System Architecture	A single system or application	Structure, components, modules, interfaces and data flow of one system; ensures it meets performance and integration needs
Enterprise Architecture (EA)	The entire organisation	Aligns business strategy, processes, data, applications and technology infrastructure across all systems; a top-down, holistic blueprint
Application Architecture	One application's internal design	How modules, layers (presentation / business / data) and components interact internally
Internet Architecture	Networked, distributed systems on the web	How data, protocols (TCP/IP, HTTP), servers and clients communicate to deliver content over the internet

⭐ **Exam tip**

Key relationship: **Enterprise Architecture is the broadest**; System/Application Architecture zoom into individual systems; Internet Architecture is the network/communication layer that connects everything. The summary line in your notes: **System < Application < Enterprise** in scope.

⚠️ **Careful**

The figure draws System and Application inside one blue region because both concern a single system's internals — in exams, keep their definitions separate as in the table.

🧠 **Yaad rakho**

Enterprise = **poori company**. System/Application = **ek system andar se**. Internet = **jodne wala network**.

📖 GFG: System vs software architecture · Enterprise architecture vs distributed system · Enterprise architect framework

2.6 Research and Practice in Information Architecture

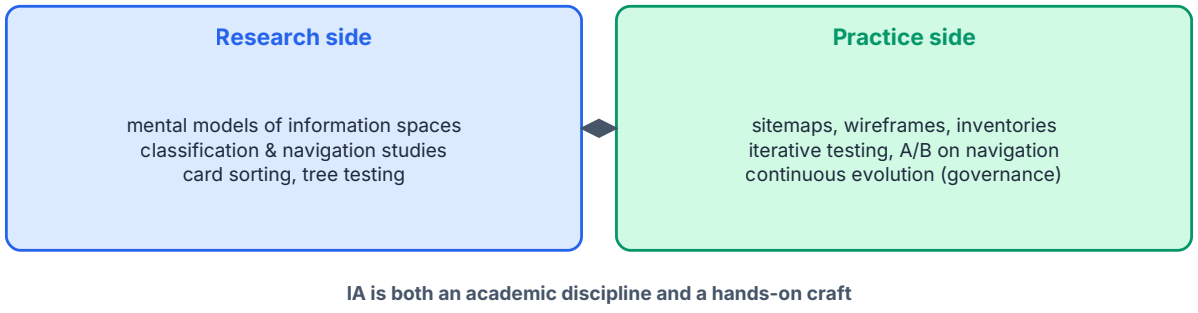


Fig. 20 — IA is both a research discipline and a hands-on practice

Research side studies	Practice side involves
How users form mental models of information spaces	Real deliverables: sitemaps, wireframes, content inventories, taxonomy documents
Effectiveness of classification/navigation schemes (controlled studies)	Iterative testing with real users (usability, A/B testing of navigation)
Card sorting and tree-testing methods to validate structures	Continuous evolution of IA as content grows (governance)

Card sorting

📘 **Definition**

Card sorting is a key practical research technique in which users group content items written on cards, revealing how they expect information to be organised.

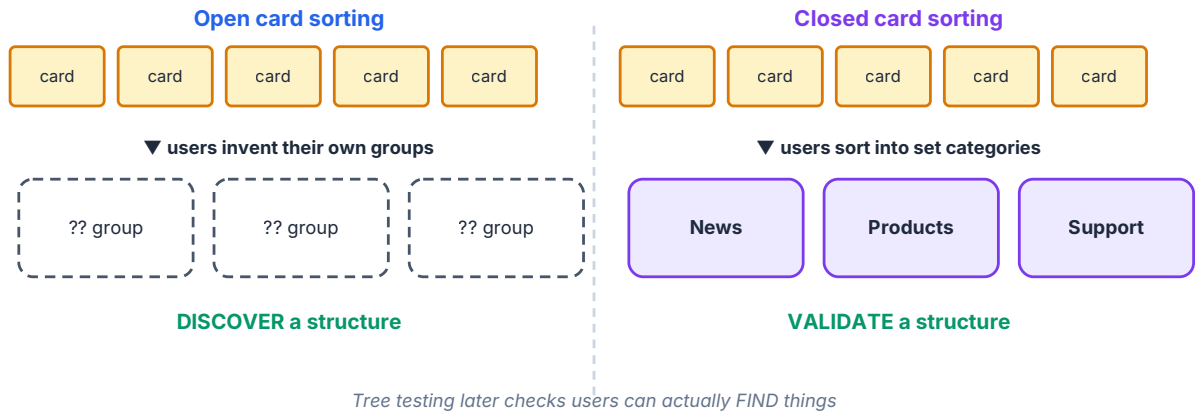


Fig. 21 — Open card sorting discovers a structure; closed card sorting validates one

Type	How	Used to
Open	Users create their own categories and group items into them	Discover a structure
Closed	Users sort items into categories that are already defined	Validate a structure

🧠 **Yaad rakho**

Open = **khud categories banao** (khojo). Closed = **diye gaye categories mein daalo** (jaancho).

❓ **Likely exam questions**

- What is Information Architecture? How is it different from UX?
- Explain the 8 principles of IA with examples.
- Describe the role and responsibilities of an information architect.
- List the areas of IA and the four systems they map onto.
- Differentiate System, Enterprise, Application and Internet architecture.
- Explain open and closed card sorting.
- Write short notes on research and practice in IA.

📖 GFG: IA — card sorting · CleverX — Mastering card sorting methods

UNIT III

Organizing Information

Organization schemes · Challenges · Websites & intranets · Cohesive systems · WWW & browser · Context & flexibility · Navigation types · Integrated & remote · Elegant navigation

3.1 Organizing Information & Organizational Challenges

Definition

Organizing information means grouping content into logical, predictable categories using an organization scheme.

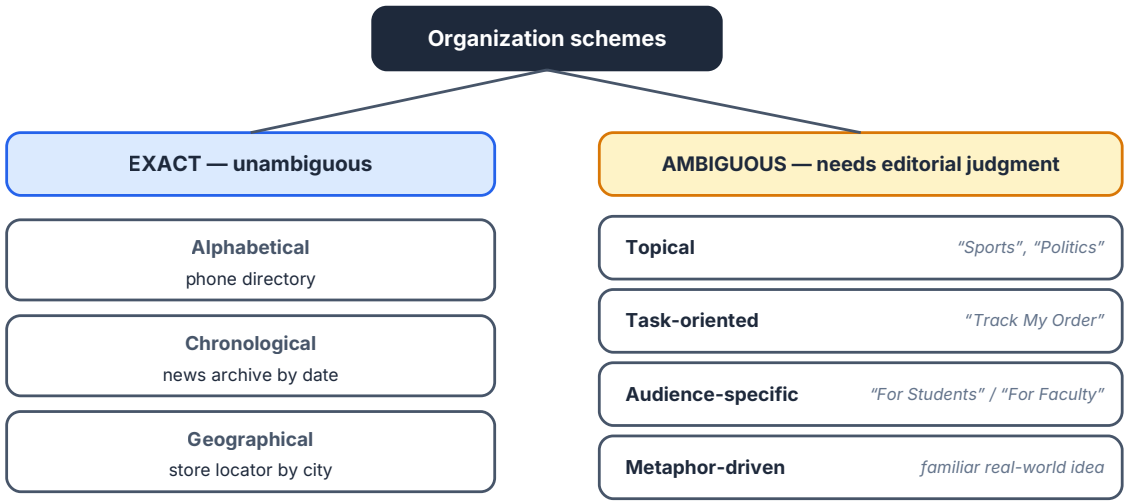


Fig. 22 — Exact schemes are unambiguous; ambiguous schemes need editorial judgment

Kind	Scheme	Example / idea
Exact unambiguous — users know where to look	Alphabetical	Phone directory
	Chronological	News archive by date
	Geographical	Store locator by city
Ambiguous editorial judgment — harder but often more useful	Topical	By subject: "Sports", "Politics"
	Task-oriented	By what the user wants to do: "Track My Order"
	Audience-specific	By who the visitor is: "For Students" / "For Faculty"
	Metaphor-driven	A familiar real-world concept as the organizing idea

Organizational challenges

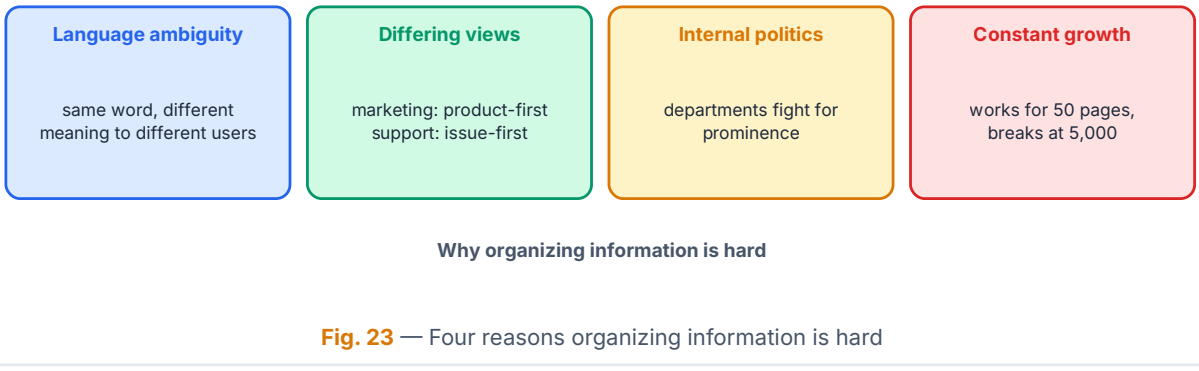


Fig. 23 — Four reasons organizing information is hard

1. **Ambiguity of language** — the same word means different things to different users (heterogeneity).
2. **Differences in perspective** — marketing wants product-first; support wants issue-first.
3. **Internal politics** — departments want prominence, creating biased structures.
4. **Constant content growth** — a scheme that works for 50 pages breaks at 5,000.

🧠 **Yaad rakho**

Exact = **A-C-G** (Alphabet, Chronology, Geography). Ambiguous = **T-T-A-M** (Topic, Task, Audience, Metaphor).

⭐ **Exam tip**

Always say **why** exact schemes are easy (no judgment needed) and **why** ambiguous ones are harder but more useful (they match how people think). Give one example each.

📖 GFG: IA — a complete guide · Big Sea — What is IA? (organizational systems)

3.2 Organizing Websites and Intranets

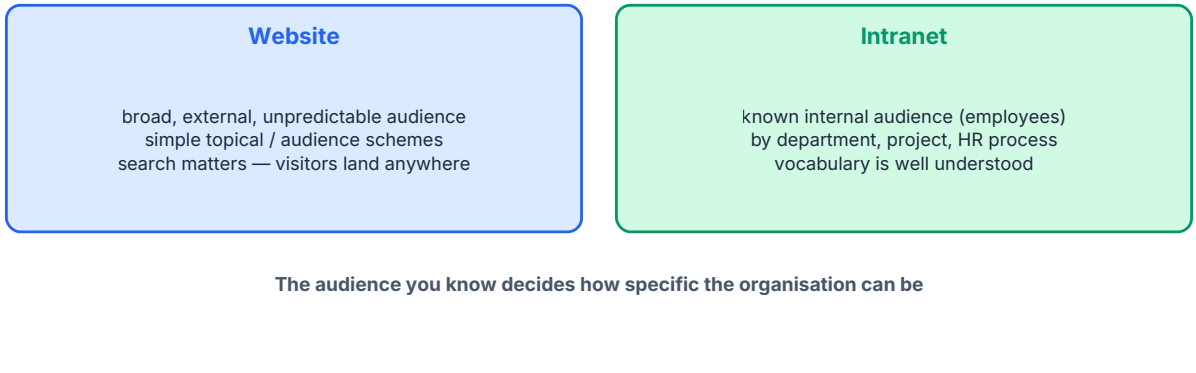


Fig. 24 — Websites serve a broad external audience; intranets serve a known internal one

- **Websites** serve a broad, external, less predictable audience → simpler, intuitive schemes (topical, audience-based); search matters because visitors may land on any page via search engines.
- **Intranets** serve a known internal audience (employees) → organization can follow internal processes, departments and workflows (department, project, HR process) because the vocabulary is well understood.

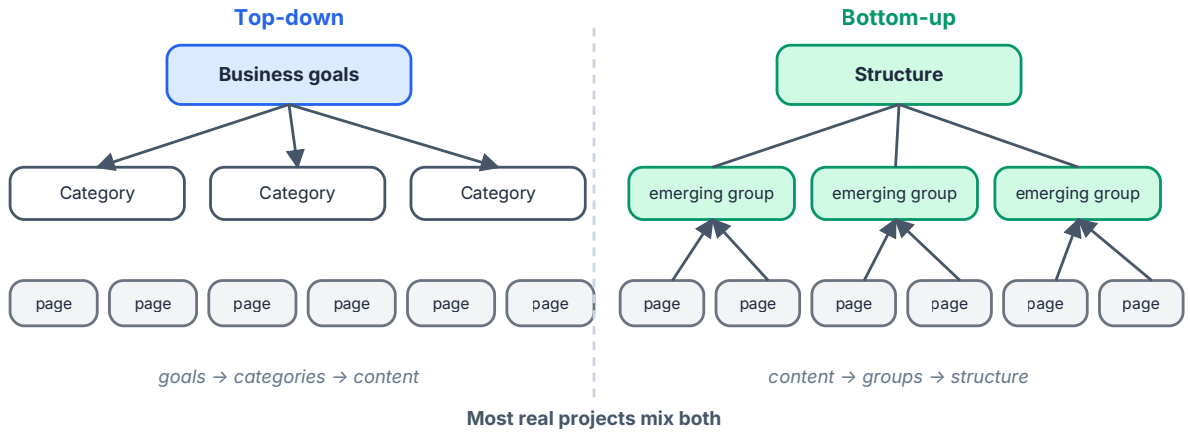


Fig. 25 — Top-down starts from goals; bottom-up starts from the content that exists

Key design decision for both: **top-down** (start from business goals and structure content into categories) vs **bottom-up** (start with the actual content and let the structure emerge). Most real projects use a mix.

🏭 **Real-life use case — A college**

The public college website is organised by **audience** ("Students", "Parents", "Alumni") and topic. The staff intranet is organised by **department** and HR process ("Leave forms", "Payroll"). *(illustration)*

3.3 Creating Cohesive Organization Systems

📘 **Definition**

A **cohesive organization system** layers multiple schemes together so users with different needs and mental models can all succeed, instead of relying on one scheme.

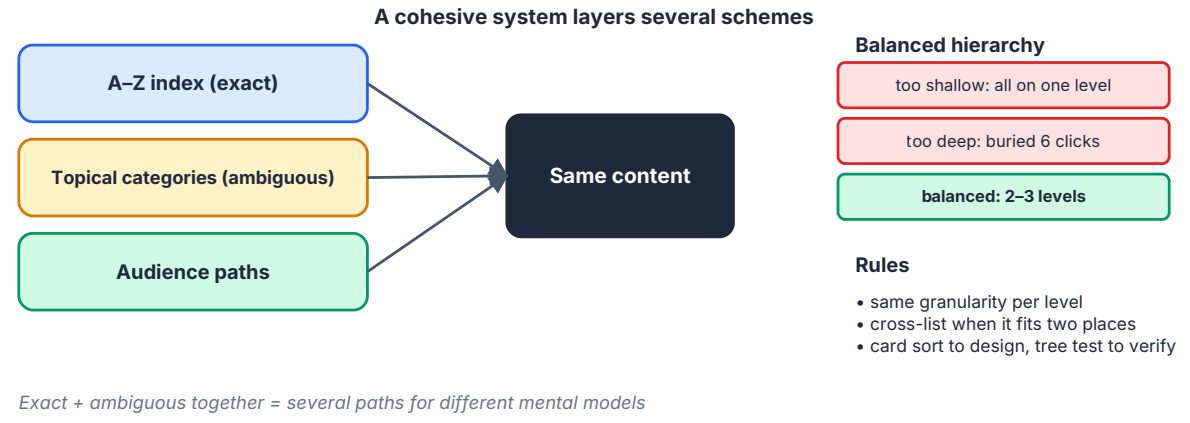
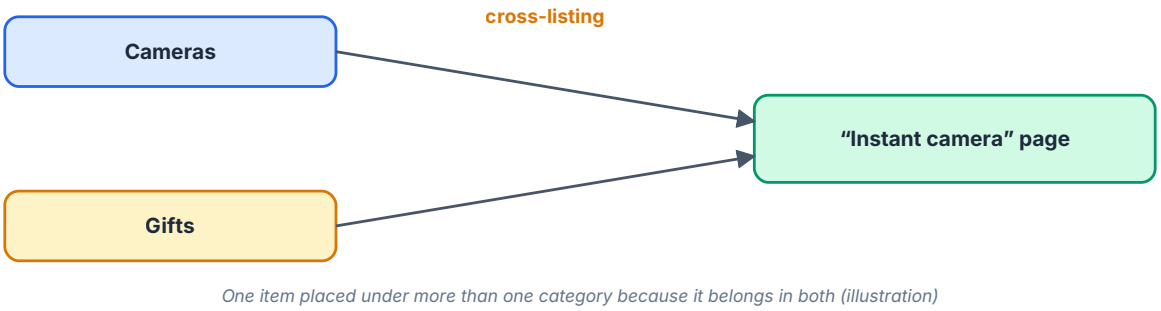


Fig. 26 — Layered schemes, a balanced hierarchy and clear rules

- Combine an **exact scheme** (A–Z index) with an **ambiguous scheme** (topical categories) → multiple paths to the same content.
- Keep the hierarchy **balanced** — not too shallow and not too deep (content buried 6 clicks down).
- Maintain **consistent granularity** — categories at one level cover roughly the same scope.
- Use **cross-listing** — the same content under more than one category when it logically belongs in several.
- Validate with **card sorting** (to design) and **tree testing** (to verify users can find things).



One item placed under more than one category because it belongs in both (illustration)

Fig. 27 — Cross-listing — one page reachable from two categories

🧠 **Yaad rakho**

Cohesive = **mix + balance + same size + cross-list + test**.

3.4 Organizing the WWW & Browser Navigation Features

📘 **Definition**

Organizing the **World Wide Web** at large is far messier than organizing one site: no central authority, constant additions and different schemes on different sites.

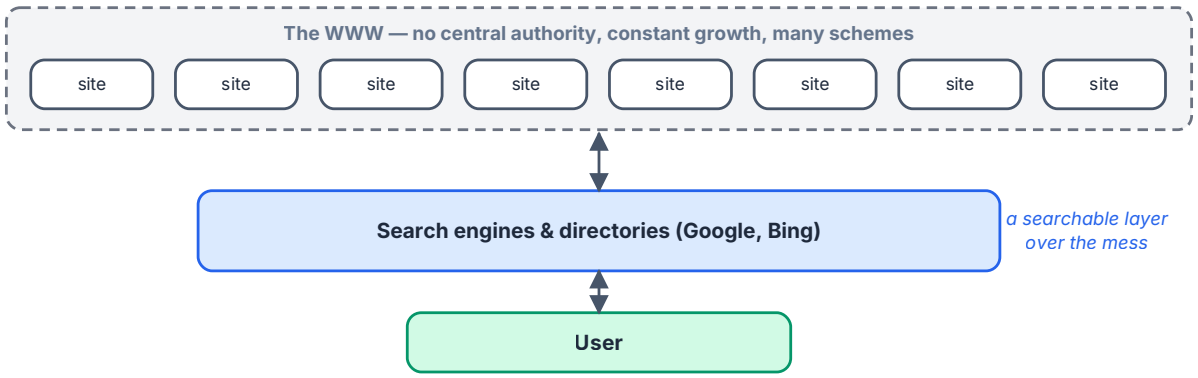


Fig. 28 — Search engines and directories impose a searchable layer over the unorganised web

That is why **search engines (Google, Bing) and directories** became essential — they impose a searchable layer over an inherently unorganised web.

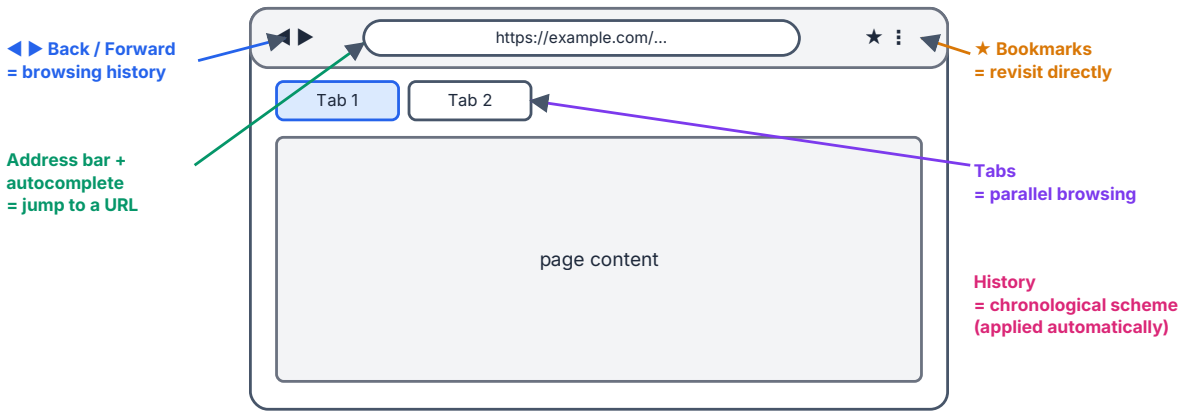


Fig. 29 — Browser features that act as the user's personal navigation aids

Browser feature	Role
Back / Forward buttons	Move through browsing history
Bookmarks / Favorites	Save and revisit important pages directly, bypassing site navigation
Address bar & autocomplete	Jump directly to a known URL
Tabs	Parallel browsing of multiple pages/sites
History	A chronological organization scheme applied automatically by the browser

💡 **Why it matters**

These browser-level tools exist precisely because the web lacks a single, consistent IA — users need their own personal navigation aids.

3.5 Building Context & Improving Flexibility

📘 **Definition**

Building context means that on whatever page a user lands, they know where they are in the larger structure, which site they are on and how to get back up. **Improving flexibility** means giving users multiple ways to reach the same content.

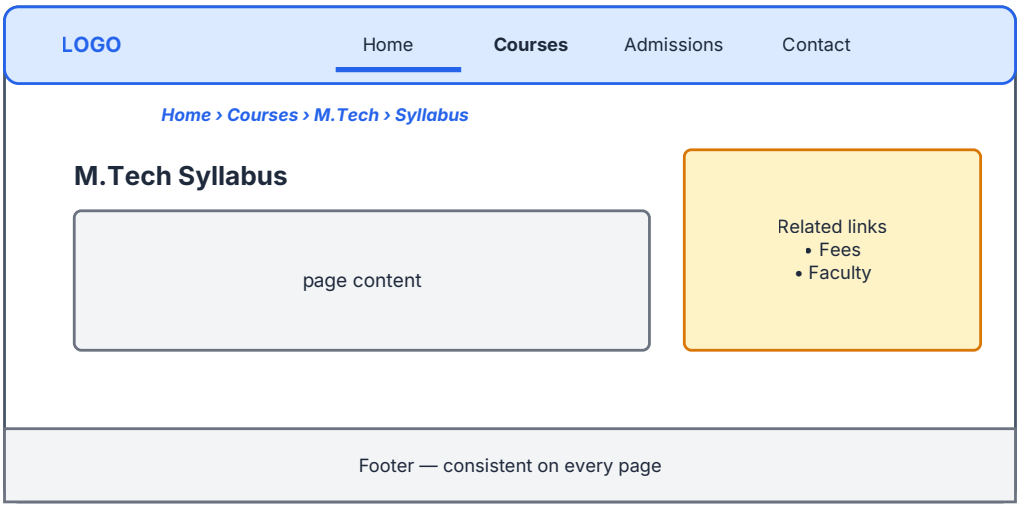


Fig. 30 — A page with the context-building elements — header, "you are here", breadcrumb, title, footer

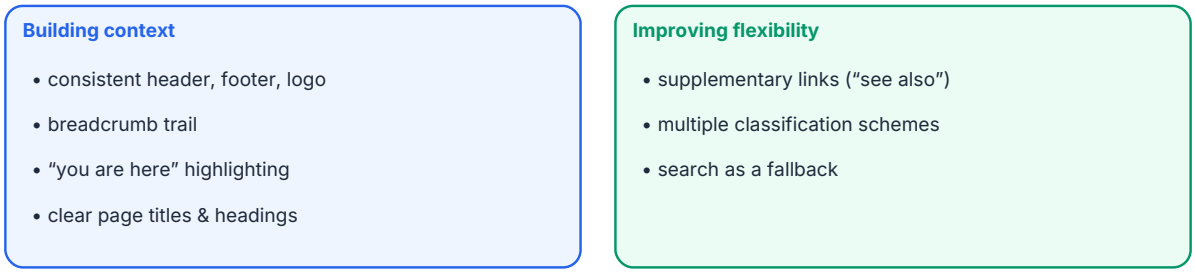


Fig. 31 — Context techniques and flexibility techniques

Context techniques	Flexibility techniques
Consistent headers, footers and logos on every page	Supplementary navigation (related links, "see also")
Breadcrumb trails (homepage → current page)	Multiple classification schemes (Principle of Multiple Classification)
"You are here" highlighting in menus	Search as a fallback when navigation fails
Page titles/headings that describe the content	

🧠 **Yaad rakho**

Context = **"Main kahan hu?"** (header, breadcrumb, highlight). Flexibility = **"Aur kitne raaste hain?"** (related links, search).

3.6 Types of Navigation Systems

📘 **Definition**

Navigation systems are the menus and links that let users move through a site. Following Rosenfeld & Morville, they are classified into four types: hierarchical, global, local and contextual.

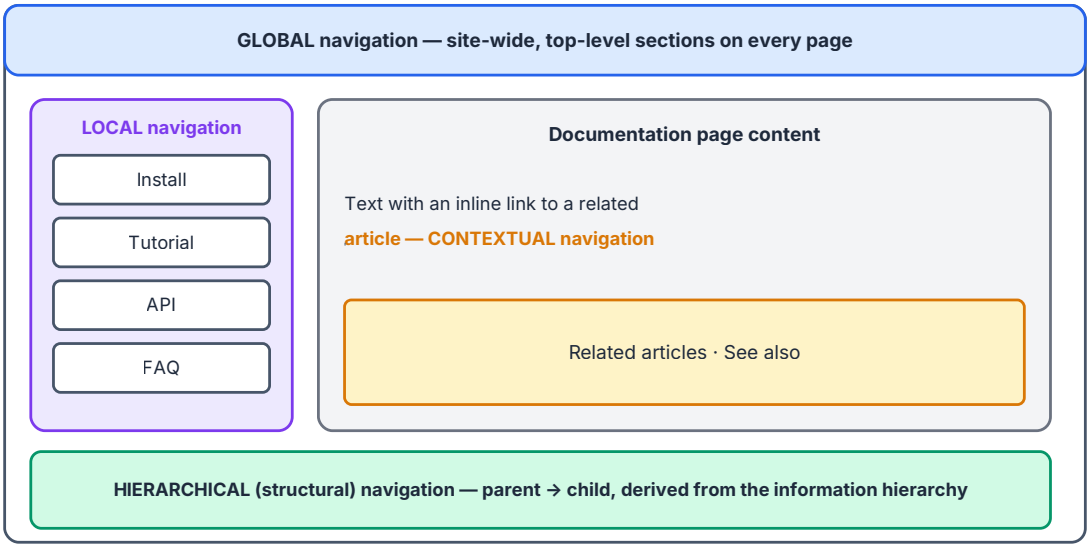


Fig. 32 — Hierarchical, global, local and contextual navigation on one page

#	Type	Meaning
1	**Hierarchical (structural)**	Primary navigation derived from the information hierarchy (parent → child categories) — the "main" navigation most sites rely on
2	**Global**	Site-wide, usually in the header, on every page — access to top-level sections from anywhere
3	**Local**	Specific to a sub-section — e.g. a sidebar menu inside a documentation section
4	**Contextual (ad hoc / embedded)**	Links inside the content connecting to related pages the hierarchy alone wouldn't surface ("related articles", inline hyperlinks)

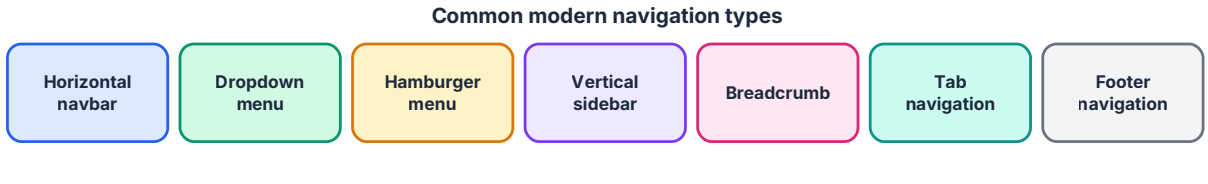


Fig. 33 — Other commonly used navigation types in modern design

Modern designs also use horizontal navbars, dropdown menus, hamburger menus, vertical sidebars, breadcrumbs, tab navigation and footer navigation.

> ⚠️ **Careful**
>
> The mock page in the figure is an illustration; in a real site the hierarchical system is the underlying structure that global and local menus are drawn from.

> 🧠 **Yaad rakho**
>
> **H-G-L-C**: Hierarchical (structure), Global (poori site), Local (ek section), Contextual (content ke andar).

Types of Navigation Systems — *Information Architecture for the World Wide Web* (O'Reilly)

3.7 Integrated & Remote Navigation Elements

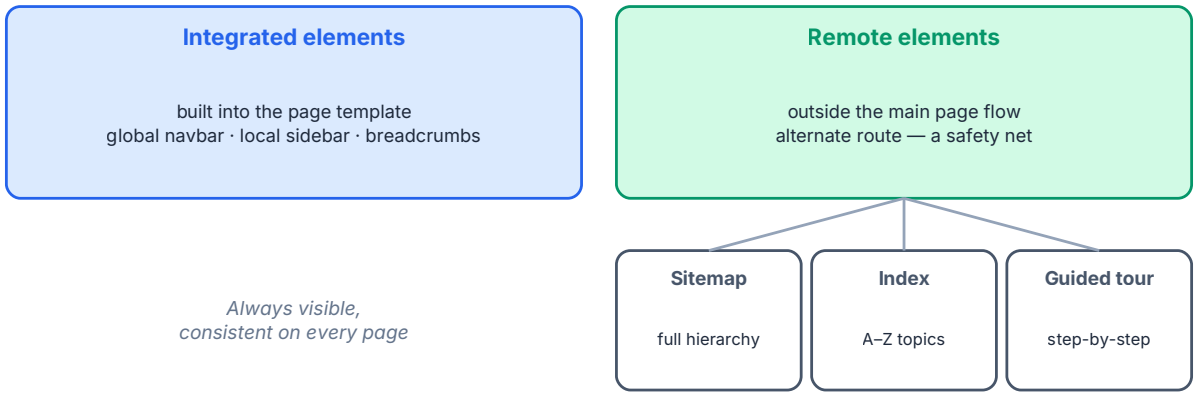


Fig. 34 — Integrated elements live in the page; remote elements sit outside the main flow

- **Integrated navigation elements** are built into the page layout as part of the main design — the global navbar, local sidebar menus and breadcrumbs inside the page template.
- **Remote navigation elements** sit outside the integrated flow as supplementary tools offering an alternate route:
  - **Sitemaps** — a page listing the whole site hierarchy, useful to users and search engines.
  - **Indexes** — an alphabetical listing of topics/keywords with links, like a book's index.
  - **Guided navigation / tours** — structured sequential paths, often for tutorials or onboarding.

💡 **Why it matters**

Remote elements are a **safety net** for users who get lost with integrated navigation alone — they provide a "big picture" or alternative access route.

🧠 **Yaad rakho**

Integrated = **page ke andar**. Remote = **page ke bahar ka backup** (sitemap, index, tour).

3.8 Designing Elegant Navigation Systems

📘 **Definition**

An **elegant navigation system** balances simplicity with completeness.

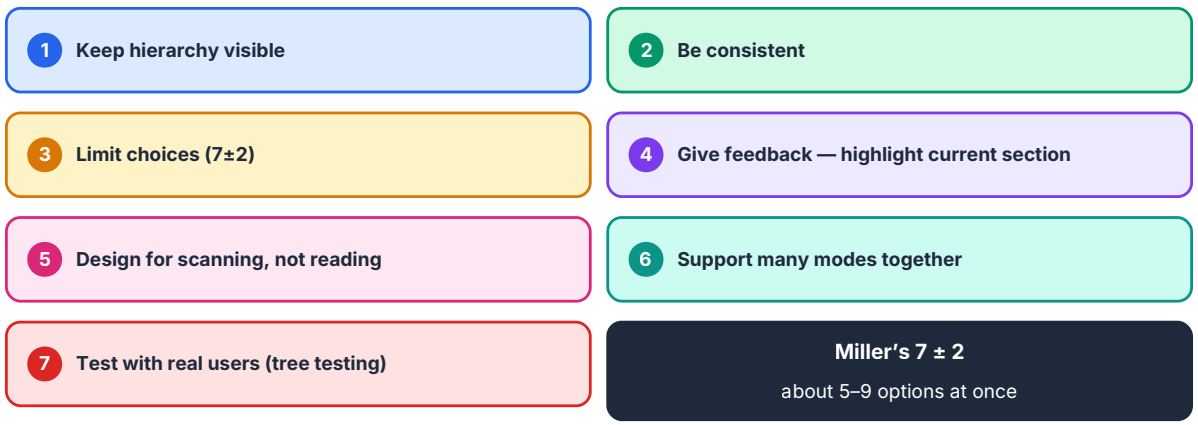


Fig. 35 — Seven guidelines for elegant navigation

1. **Keep the hierarchy visible** — global, local and contextual navigation must not contradict the content hierarchy.
2. **Be consistent** — same placement, styling and behaviour everywhere.
3. **Limit choices per screen** — Miller's "7±2": about 5–9 options at once.
4. **Provide feedback** — highlight the current section ("where am I").
5. **Design for scanning, not reading** — clear, front-loaded labels.
6. **Support multiple navigation modes together** — hierarchical + global + local + contextual + search in harmony.
7. **Test with real users** — tree testing, to confirm navigation helps people finish tasks.

> ⭐ **Exam tip**
>
> If asked to "design/critique a navigation system", structure the answer on these seven points and name **Miller's 7±2** and **tree testing** — the two most-quoted items.

> ❓ **Likely exam questions**
>
> - Explain exact and ambiguous organization schemes with examples.
> - What are the organizational challenges in IA?
> - Compare organizing a website and an intranet. Top-down vs bottom-up?
> - How do you create a cohesive organization system?
> - Why is the WWW hard to organise? Explain browser navigation features.
> - Explain how to build context and improve flexibility.
> - Explain the four types of navigation systems.
> - Integrated vs remote navigation elements.
> - Discuss guidelines for designing elegant navigation systems.

📖 GFG: IA — a complete guide · Loop11 — A beginner's guide to IA in UX

UNIT IV

Labeling Systems & Search

Labeling systems | Label types | Creating labels | Fine-tuning | Non-representational labels | Double challenge | How users search | Search interface | Reference interview | To search or not

4.1 Labeling Systems — "Not Labels"

📘 **Definition**

A **labeling system** is the set of words, phrases and icons used to represent chunks of content — navigation items, headings, links — applied as one consistent vocabulary across a site.

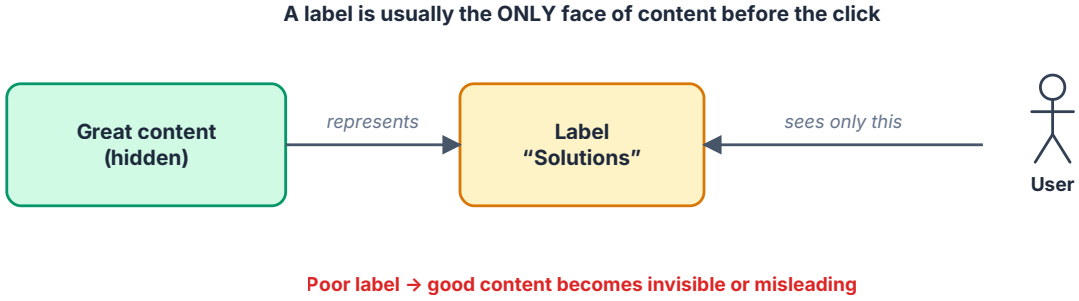


Fig. 36 — The label is often the only representation of content a user sees before clicking

The syllabus says **"labeling systems, not labels"** because a labeling system is not one label; it is the overall, consistent vocabulary across the whole site.

💡 **Why it matters**

A poor label can make excellent content invisible or misleading. Good labeling **reduces cognitive load** and lets users predict what they will find before they click.

🧠 **Yaad rakho**

Label = **content ka chehra**. Ek label nahi — poora **system** consistent hona chahiye.

Information Architecture: A Practical Guide (labeling systems section)

4.2 Types of Labeling Systems

📘 **Definition**

Labels appear on websites in two formats — **textual** and **iconic** — and serve five typical roles.

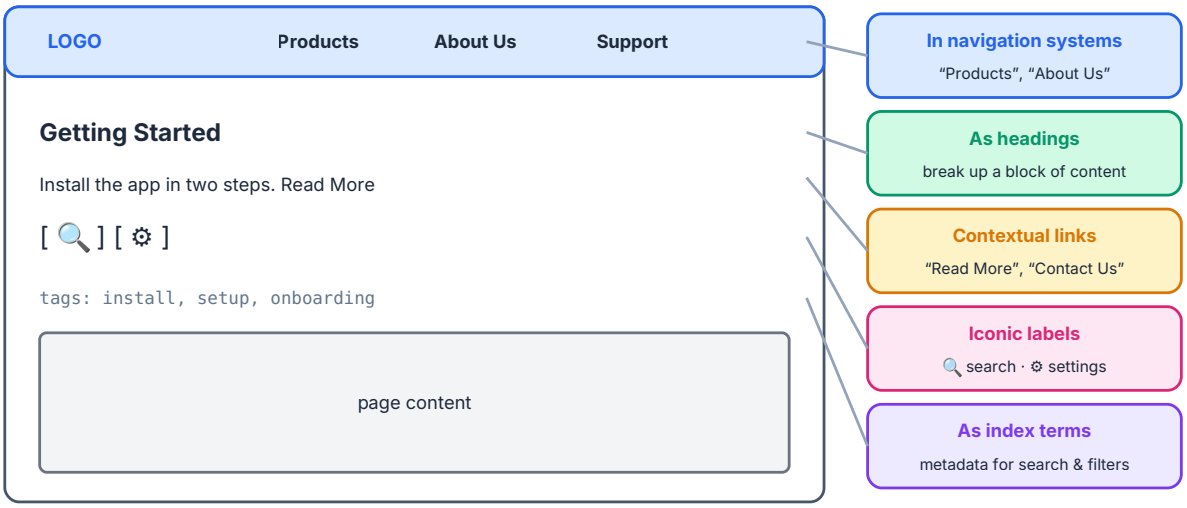


Fig. 37 — The five roles labels play on a page

#	Role	Example
1	**Contextual links**	Anchor text of a hyperlink — "Read More", "Contact Us"
2	**Within navigation systems**	Menu / nav item names — "Products", "About Us"
3	**Headings**	Text that identifies and breaks up a block of content
4	**Index terms**	Keywords/metadata behind the scenes for search and filtering (not always visible)
5	**Iconic labels**	Icons instead of / alongside text — magnifying glass = search, gear = settings

🧠 **Yaad rakho**

5 roles: "**Link, Nav, Heading, Index, Icon**".

Types of Labeling Systems — *Information Architecture for the World Wide Web* (O'Reilly) · Information Architecture: A Practical Guide

4.3 Creating Effective Labeling Systems

📘 **Definition**

Effective labels use consistent, user-tested terminology within a narrow content domain, following style rules decided once and applied everywhere.

1. **Narrow the scope** — one content domain, consistent terminology; don't borrow inconsistent terms from many sources.
2. **Develop consistent style rules** — case, verb form, length.
3. **Use familiar, user-tested terminology** — from real user language (card sorting, search logs), not internal jargon.
4. **Choose good sources** for labels.

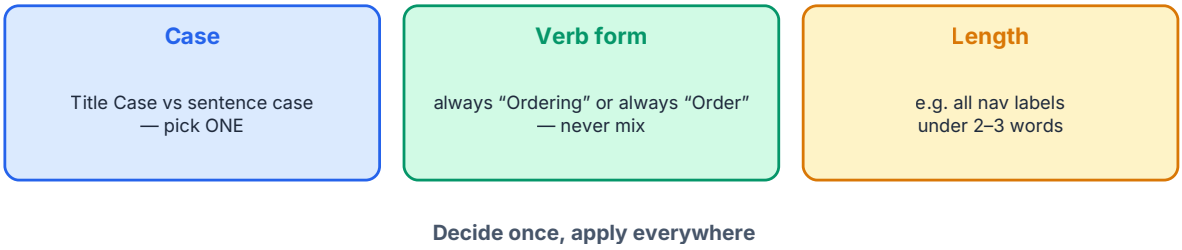


Fig. 38 — Style rules decided once — case, verb form and length

Examples from the notes: case (Title Case vs sentence case); verb form (always "Ordering" vs always "Order" for gerund-style labels); length limits (all nav labels under 2–3 words).

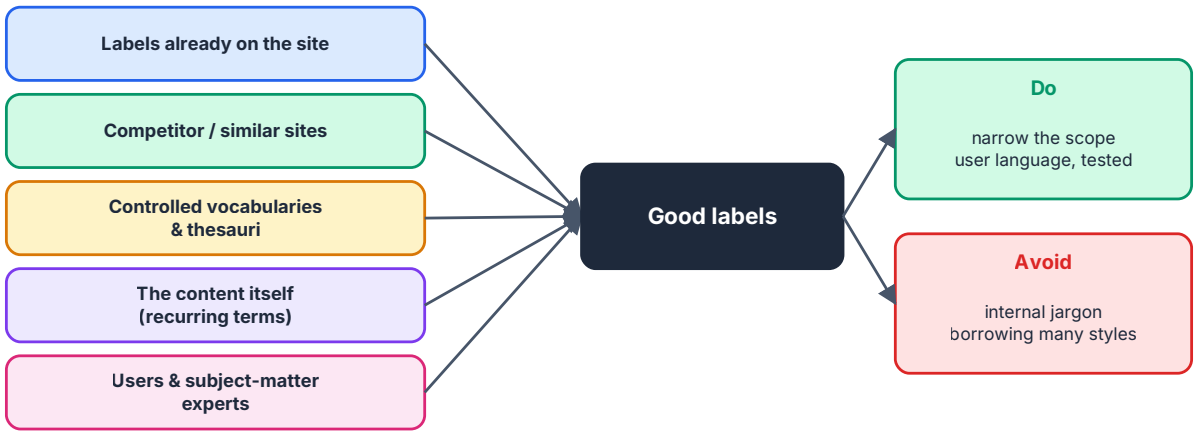


Fig. 39 — Five sources of good labels

Source	What you get
Labels already on the site	Existing conventions
Competitor / other sites in the domain	Familiar terms users already know
Controlled vocabularies & thesauri	Industry-standard term lists
The content itself	Recurring terms extracted from the text
Users and subject-matter experts	Terms from interviews and testing

🏭 Real-life use case — Nav labels that work

A college site agrees: Title Case, noun phrases, max two words — "Admissions", "Academic Calendar", "Contact Us" — and never mixes in "Get to know us" or "Click here to apply". *(illustration)*

⭐ Exam tip

Remember the three style rules as **case, verb form, length** and the five sources as **site, competitors, thesauri, content, users/experts**.

Blog: Information Architecture — labelling table exercise (Medium) · O'Reilly IA book

4.4 Fine-Tuning the Labeling System

📘 **Definition**

Fine-tuning means refining a drafted labeling system iteratively with testing, audits and analytics.

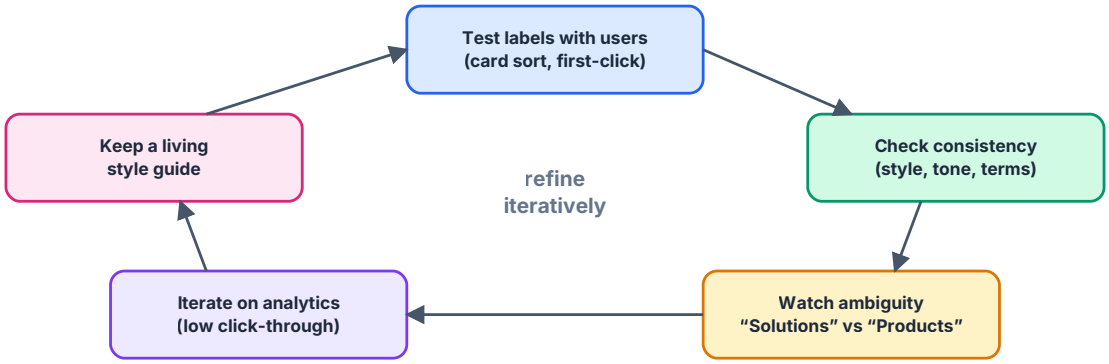


Fig. 40 — Fine-tuning is a loop, not a one-off check

- **Test labels with real users** — card sorting (do users group items as the labels suggest?) and **first-click testing** (do they click the right label?).
- **Check consistency** — audit labels for style, tone and terminology drift.
- **Watch for ambiguity** — a label clear internally may confuse outsiders (e.g. "Solutions" vs "Products").
- **Iterate on analytics** — low click-through or high bounce rates signal labels to revise.
- **Maintain a living style guide / glossary** of approved labels.

🧠 **Yaad rakho**

Fine-tune = **Test, Check, Dekho ambiguity, Analytics, Glossary**.

4.5 Non-Representational Labels & The Double Challenge

📘 **Definition**

Non-representational labeling systems use arbitrary codes, numbers or internal names that carry no inherent meaning to outsiders.

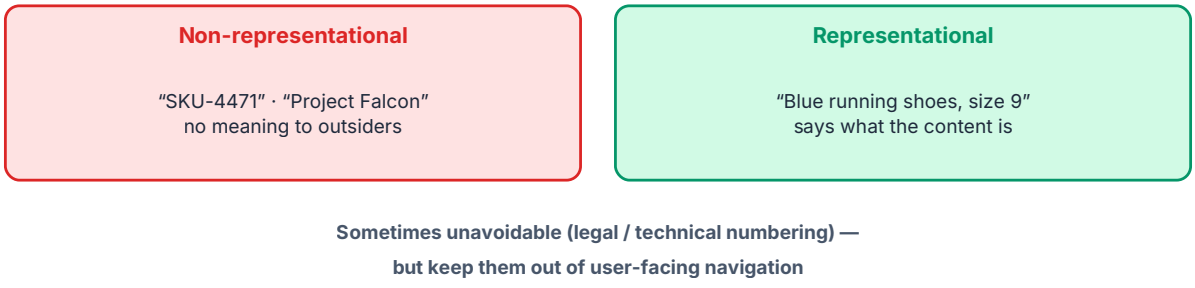


Fig. 41 — A code tells outsiders nothing; a descriptive label does

Examples: product codes like "SKU-4471" or internal project codenames. They are sometimes unavoidable (legal/technical document numbering) but should be **minimised in user-facing navigation**.

The Double Challenge

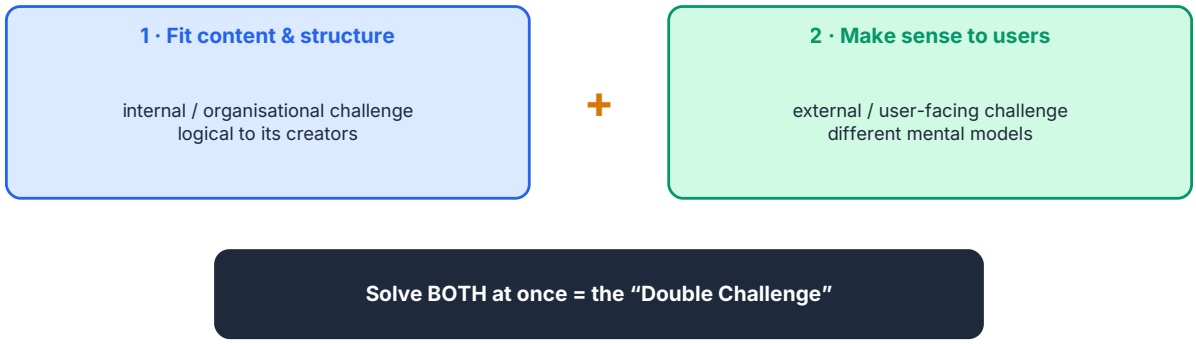


Fig. 42 — Solve the internal and the external problem at the same time

1. Design a labeling system that **fits the content and structure** (internal, organizational challenge).
2. Make sure labels **make sense to real end users**, whose mental models may differ from the builders' (external, user-facing challenge).

💡 **Why it matters**

A system can be perfectly logical to its creators and still fail with real users — balancing internal logic against outside comprehension is what makes labeling genuinely difficult.

🧠 **Yaad rakho**

Double challenge = **andar ka logic + bahar ka user** — dono ko ek saath khush rakhna.

4.6 Searching a Website — Understanding How Users Search

📘 **Definition**

Before designing search you must understand **why and how users search**; their needs fall into four search-behaviour patterns.

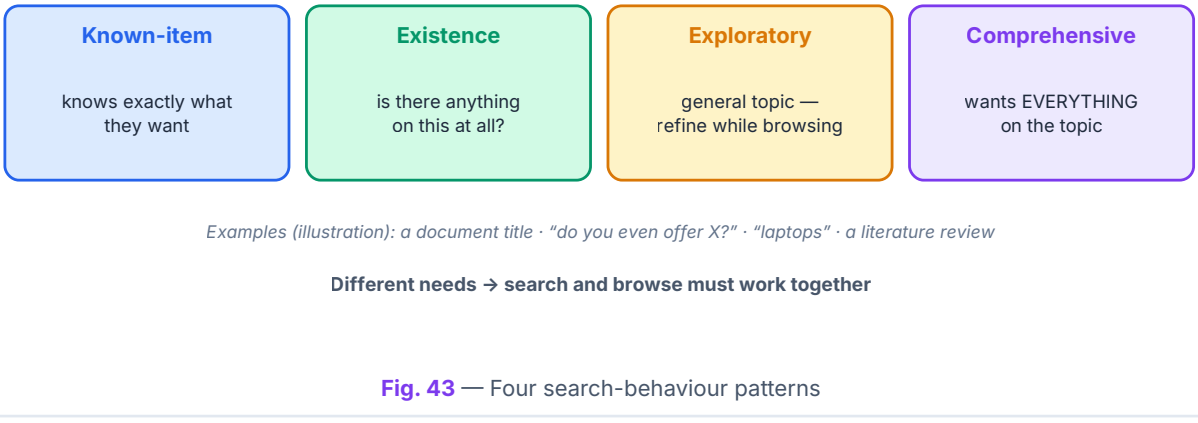


Fig. 43 — Four search-behaviour patterns

Pattern	Meaning
Known-item	User knows exactly what they want (e.g. a specific document title)
Existence	User isn't sure the information exists and searches to find out
Exploratory	General topic in mind, unsure what exactly; refines while browsing results
Comprehensive (research)	User wants everything related to a topic, not just one answer

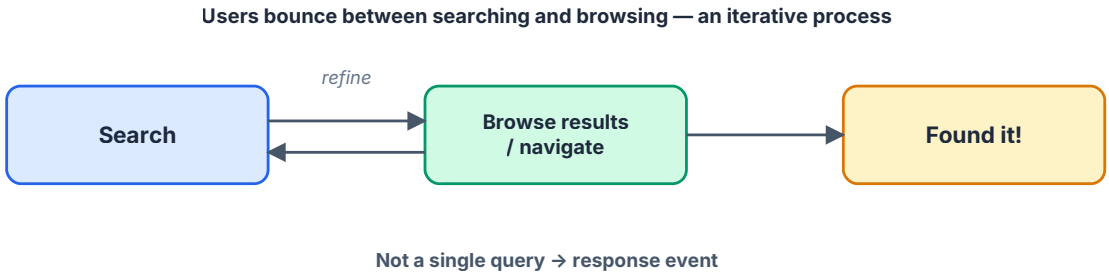


Fig. 44 — Search and browse feed each other in an iterative loop

Because needs differ so much, **search and browse are deeply integrated** — users bounce between them several times; it is an iterative process, not a single query-response.

🧠 **Yaad rakho**

Search ke 4 prakar: "**K-E-E-C**" — Known, Existence, Exploratory, Comprehensive.

4.7 Designing the Search Interface

⭐ **Exam tip — Frequently asked as a design / critique question**

Learn the eight guidelines below and the three-component architecture at the end.

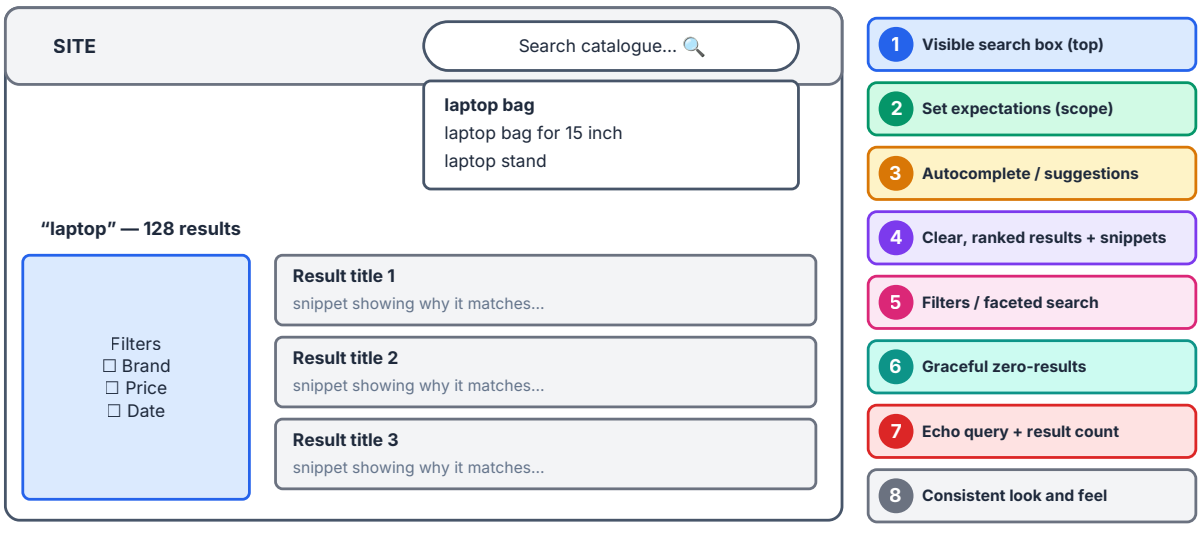


Fig. 45 — A search page with the eight guidelines listed alongside

1. **Visible, easy-to-find search box** — typically top-right or top-centre.

2. **Set expectations** — show what is searched (whole site or a section) via placeholder text or scoped search.
3. **Autocomplete / suggestions** — help refine queries, reduce errors.
4. **Display results clearly** — relevance-ranked results with useful snippets, not raw titles.
5. **Filtering / faceted search** — narrow by category, date, price.
6. **Handle zero-result queries gracefully** — spelling corrections or related content, not a blank page.
7. **Feedback on the query** — echo what was searched and the number of results.
8. **Consistent look and feel** with the rest of the site.

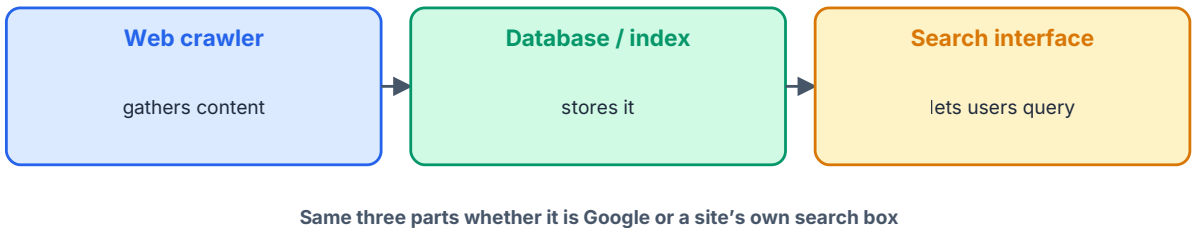


Fig. 46 — The three components of any search engine

Technically a search engine/interface has three components: **Web Crawler** (gathers content), **Database/Index** (stores it) and **Search Interface** (lets users query it) — the same whether Google or a site's internal search box.

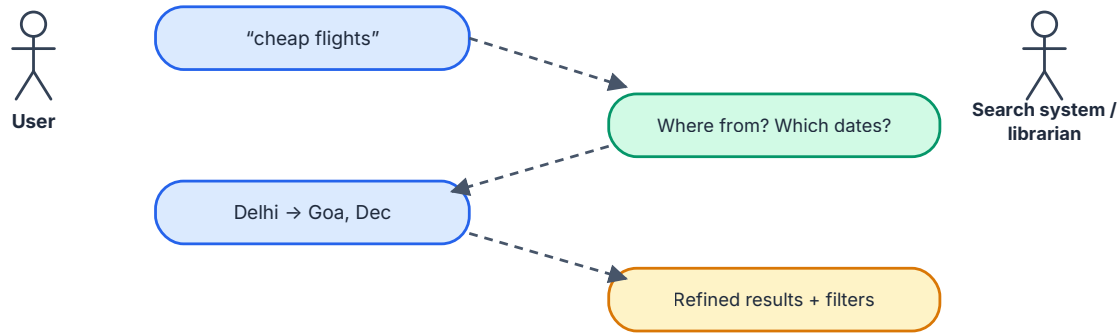
⚠️ **Careful**

The mock page in the figure is an illustration of the guidelines; the numbers in the list on the right match the numbered list above.

4.8 The Reference Interview & Indexing the Right Stuff

📘 **Definition**

The **reference interview** is a library-science concept: rather than assuming what the user wants from the first query, clarify their intent — through query-refinement suggestions, filters and clarifying prompts — like a librarian asking follow-up questions.



Clarify intent before pointing to a resource (flight example is an illustration)

Fig. 47 — Clarifying intent before answering

Indexing the right stuff

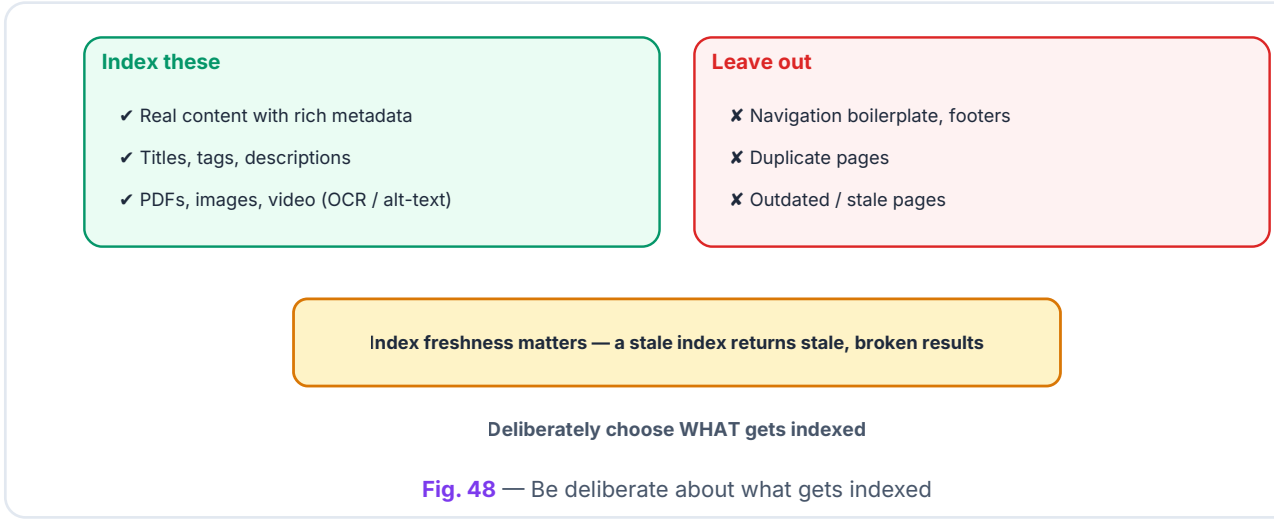


Fig. 48 — Be deliberate about what gets indexed

- Not all content should be searchable — exclude navigation boilerplate, footers and duplicate pages.
- **Metadata** (titles, tags, descriptions) should be rich enough for accurate matching.
- **Index freshness** matters — outdated indexes return outdated or broken results.
- Different content types (PDFs, images, video) need different strategies, e.g. OCR or alt-text for non-text content.

🧠 **Yaad rakho**

Reference interview = **librarian jaisa sawaal**: "Aap ko asal mein kya chahiye?" Indexing = **sahi cheez hi index karo, taaza rakho**.

4.9 To Search or Not to Search

📘 **Definition**

A critical, often-overlooked IA decision: **search is not always the right solution**.

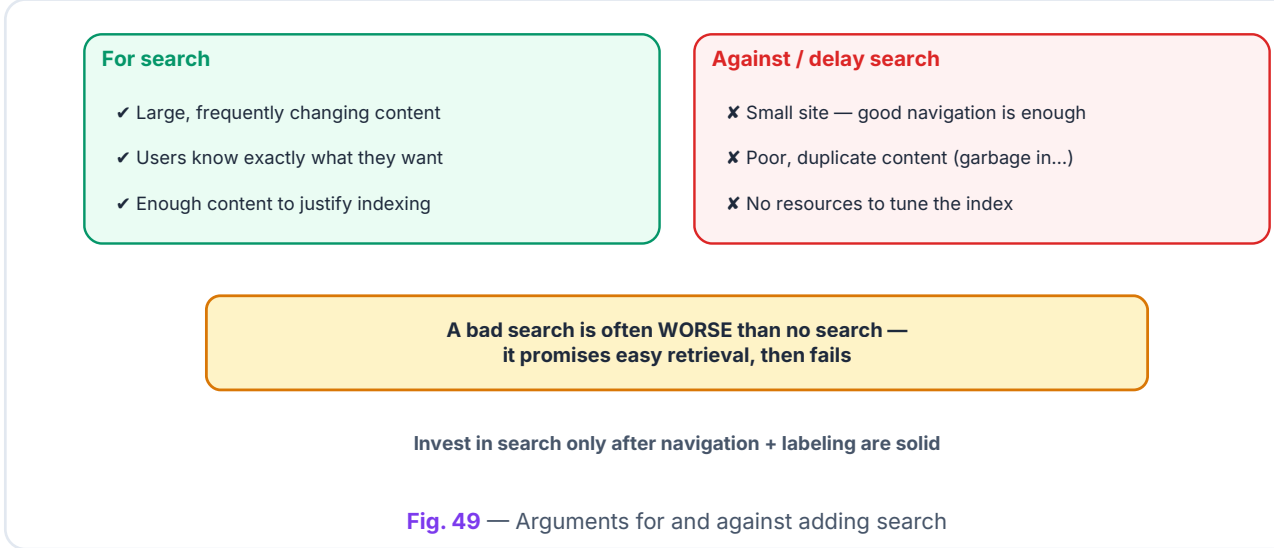


Fig. 49 — Arguments for and against adding search

For including search	Against (or for delaying) search
Large, frequently changing content	Small sites — good navigation may suffice; poor search can hurt more than none
Users often know exactly what they want (known-item)	Poor / inconsistent content — garbage in, garbage out
Enough content volume to make indexing worthwhile	Not enough resources to maintain the index and tune relevance

⭐ **Exam tip — Rule of thumb**

A **bad search experience is often worse than no search**, because it creates a false expectation of easy retrieval that then fails. Invest in search only once navigation and labeling are solid and content volume / change frequency justify it.

❓ **Likely exam questions**

- What is a labeling system? Why "labeling systems, not labels"?
- Explain the types of labels with examples.
- How do you create effective labels? List sources of labels.
- How do you fine-tune a labeling system?
- What are non-representational labels? Explain the Double Challenge.
- Explain how users search — the four search patterns.
- Discuss guidelines for designing a good search interface. Name the three components of a search engine.
- What is a reference interview? What does "indexing the right stuff" mean?
- "To search or not to search" — discuss.

📖 GFG: IA — a complete guide (card sorting / testing sections) · Information Architecture: A Practical Guide

UNIT V

The Open Group Architecture Framework (TOGAF)

TOGAF 9 | Management overview | Components | ADM | ADM phases | Deliverables | Enterprise Continuum | Governance | Views & viewpoints | Building blocks

5.1 Introduction to TOGAF 9

📘 **Definition**

TOGAF (The Open Group Architecture Framework) is the world's most widely adopted **Enterprise Architecture (EA) framework**, providing a structured approach for designing, planning, implementing and governing an enterprise's IT architecture so it stays aligned with business goals.

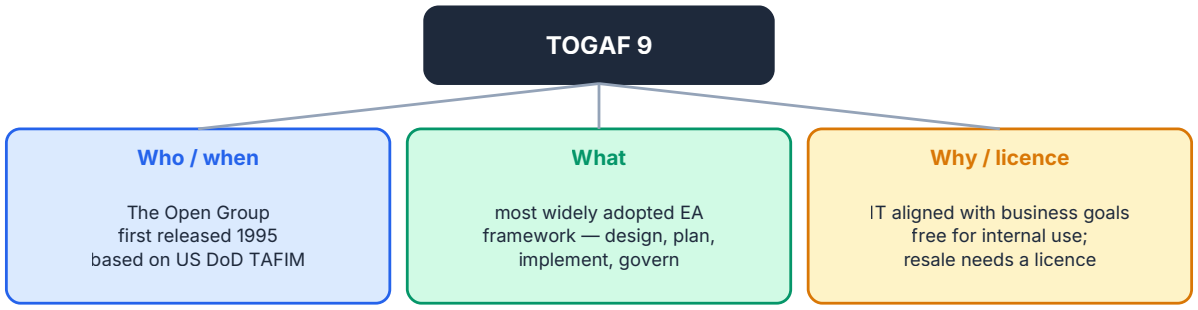


Fig. 50 — Key facts about TOGAF

- Developed and maintained by **The Open Group**; first released **1995**, based on the U.S. DoD's **TAFIM** framework.
- Models enterprise architecture at four levels — **Business, Data, Application, Technology (BDAT)**.
- Adopted by a majority of Fortune 500 and Global 50 companies.
- Free for internal use within an organisation; commercial/resale use needs Open Group licensing.

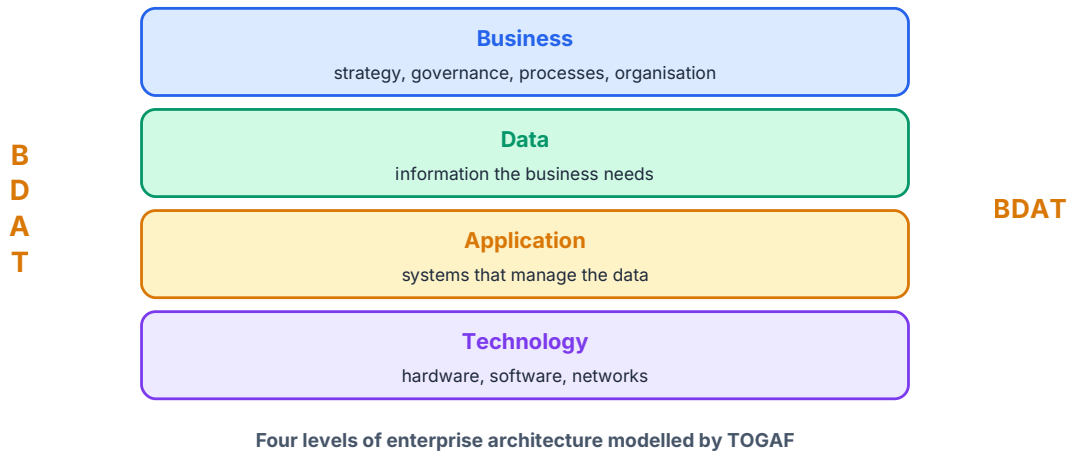


Fig. 51 — The four BDAT levels

🧠 **Yaad rakho**

TOGAF = **BDAT** — Business, Data, Application, Technology. 1995, The Open Group, TAFIM se.

TOGAF — official page, The Open Group

5.2 TOGAF 9 Management Overview

📘 **Definition**

At management level TOGAF solves a business problem: without a common framework every EA project reinvents its own process, vocabulary and deliverables, making projects hard to compare, govern or reuse.

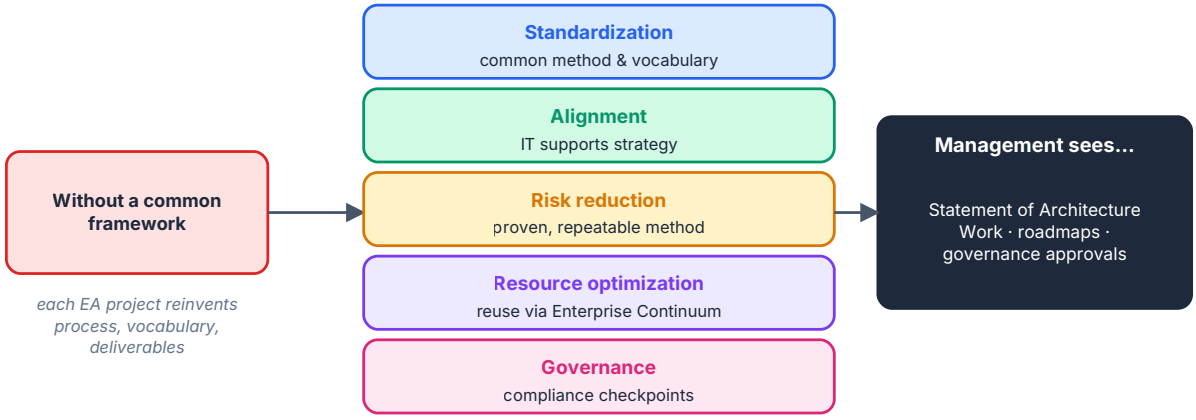


Fig. 52 — The five management-level benefits of TOGAF

Benefit	Meaning
Standardization	Common method and vocabulary for all architecture work
Alignment	IT investment directly supports business strategy
Risk reduction	Proven, repeatable method lowers the risk of failure or missed requirements
Resource optimization	Avoids duplicated effort by reusing architecture assets (Enterprise Continuum, Sec. 5.7)
Governance	Checkpoints to keep work compliant with principles and standards (Sec. 5.8)

Management typically engages with TOGAF through its **outputs** — the Statement of Architecture Work, architecture roadmaps and governance approvals — not the detailed technical artefacts.

> 🧠 **Yaad rakho**
>
> Management ko kya milta hai: "**S-A-R-R-G**" — Standardization, Alignment, Risk reduction, Resource optimization, Governance.

What is TOGAF? An EA framework for aligning technology to business — CIO.com

5.3 TOGAF 9 Components

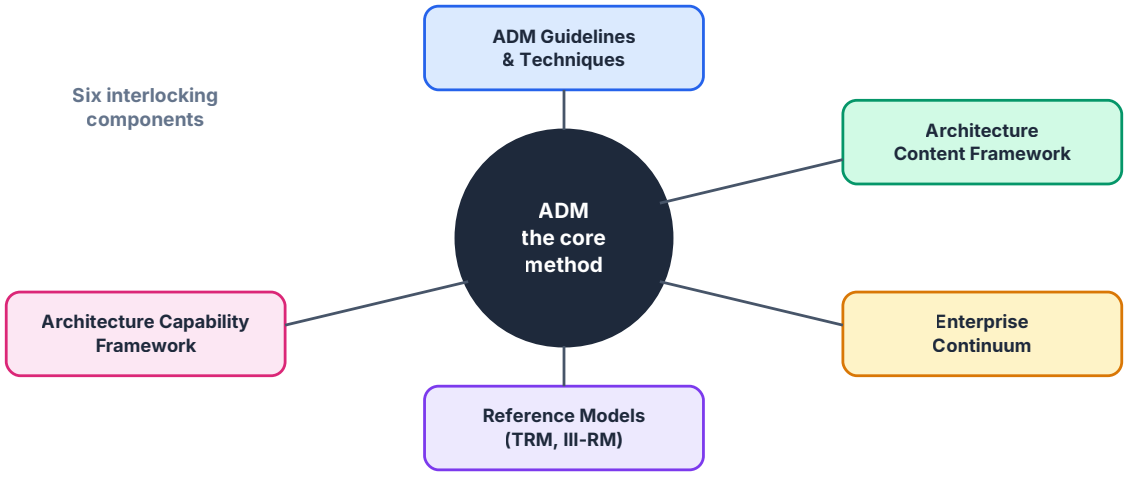


Fig. 53 — The six interlocking components of TOGAF, with the ADM at the core

#	Component	Role
1	**Architecture Development Method (ADM)**	The core, step-by-step, iterative method for developing an enterprise architecture (Sec. 5.4–5.5)
2	**ADM Guidelines & Techniques**	Guidance for adapting the ADM (e.g. security architecture, agile projects)
3	**Architecture Content Framework**	Structure of deliverables, artefacts and building blocks produced during the ADM
4	**Enterprise Continuum**	Categorisation scheme/repository classifying assets from generic to organisation-specific (Sec. 5.7)
5	**TOGAF Reference Models**	Technical Reference Model (TRM) and Integrated Information Infrastructure Reference Model (III-RM) — pre-built reference architectures
6	**Architecture Capability Framework**	Organisational structures, roles, skills and processes needed to operate and sustain the architecture practice

> ⚠️ **Careful**
>
> The figure places the ADM at the centre only to show its importance; in the notes all six are listed as equal, interlocking components.

5.4 Introduction to the ADM

Definition

The **Architecture Development Method (ADM)** is the heart of TOGAF — a proven, iterative, cyclical process for developing and managing the whole lifecycle of an enterprise architecture.

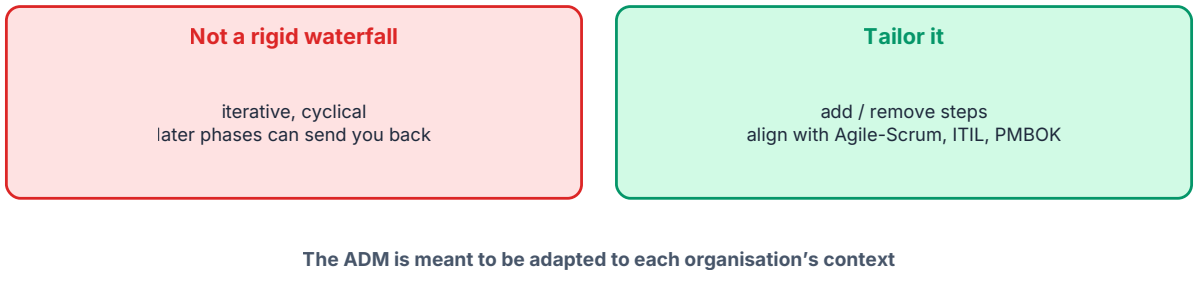


Fig. 54 — The ADM is iterative and meant to be tailored

- It is **not** a rigid waterfall; organisations tailor it — adding/removing steps and aligning with frameworks like Agile-Scrum, ITIL or PMBOK.
- The cycle revolves around a central hub, **Requirements Management**, which continuously feeds validated requirements into and out of every other phase so the architecture stays grounded in real business needs.

The TOGAF Standard 9.2 — official ADM documentation, The Open Group

5.5 ADM Phases

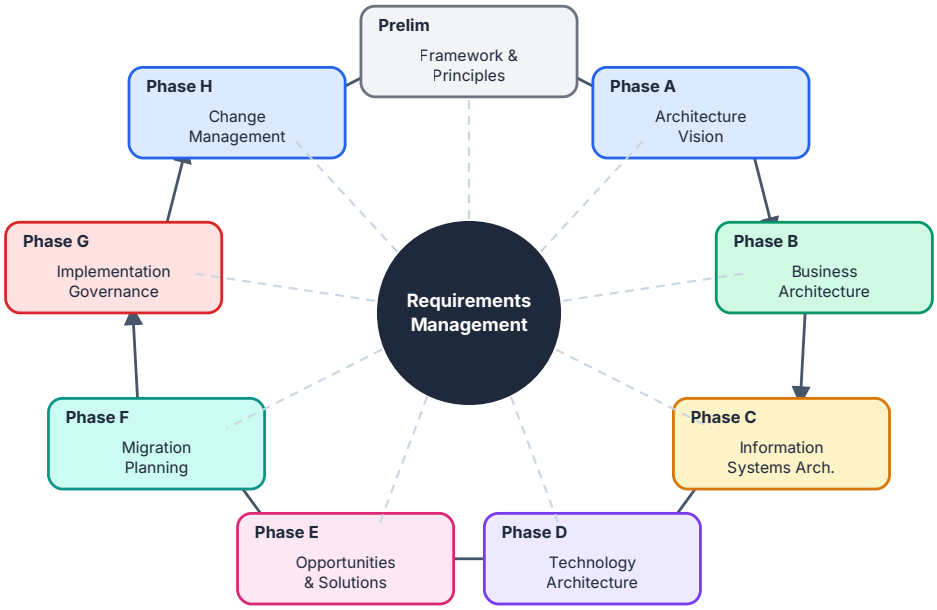
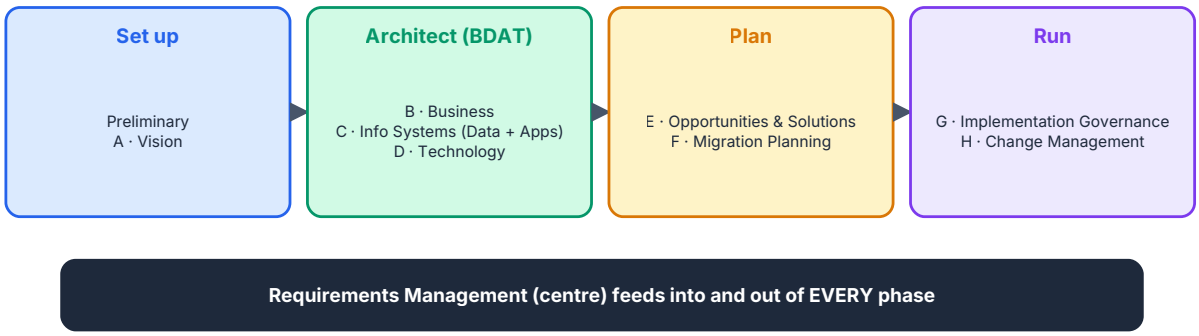


Fig. 55 — The ADM cycle — Preliminary and Phases A to H around the central Requirements Management

Phase	Name	Purpose
Preliminary	Framework & Principles	Prepares the organisation: defines architecture principles and scope, tailors the ADM before the cycle begins
Phase A	Architecture Vision	Defines scope, identifies stakeholders, creates a high-level vision, secures approval to proceed
Phase B	Business Architecture	Target business architecture — strategy, governance, processes, organisation
Phase C	Information Systems Architecture	Target Data and Application architectures — what information is needed and which applications manage it
Phase D	Technology Architecture	Target technology architecture — hardware, software, networks supporting applications and data
Phase E	Opportunities & Solutions	Identifies major implementation projects, groups them into a roadmap, evaluates build-vs-buy
Phase F	Migration Planning	Prioritises projects, assesses cost/benefit/risk, finalises the implementation and migration plan
Phase G	Implementation Governance	Architectural oversight of implementation — projects conform to the architecture
Phase H	Architecture Change Management	Monitors the running architecture, manages change requests, decides whether a new ADM cycle is needed
Requirements Management	(central, continuous)	Manages architecture requirements throughout, feeding every other phase



The grouping is a revision aid — the notes list the phases individually

Fig. 56 — A revision grouping of the ADM phases

⭐ Exam tip — Memory tip from the notes

Preliminary → A (Vision) → B (Business) → C (Info Systems: Data + Apps) → D (Technology) → E (Opportunities) → F (Migration) → G (Governance) → H (Change Mgmt) → back to Requirements Management (centre).

⚠️ Careful — Counting the phases

Your notes say the ADM has "10 phases". The table lists the Preliminary phase + Phases A–H (**9**) plus the central Requirements Management, which makes **10** items. Write it as "Preliminary, A to H around central Requirements Management" to be safe.

🧠 **Yaad rakho**

"**P-A-B-C-D-E-F-G-H**" — aur beech mein **R**equirements Management. Vision → Business → Info Systems → Technology → Opportunities → Migration → Governance → Change.

The TOGAF Standard 9.2 — Phase A: Architecture Vision (The Open Group) · KnowledgeHut — 10 Phases of TOGAF 9.2 ADM explained

5.6 ADM Deliverables

📘 **Definition**

Deliverables are the formal, tangible work products each ADM phase produces — reviewed, approved and stored in the architecture repository.

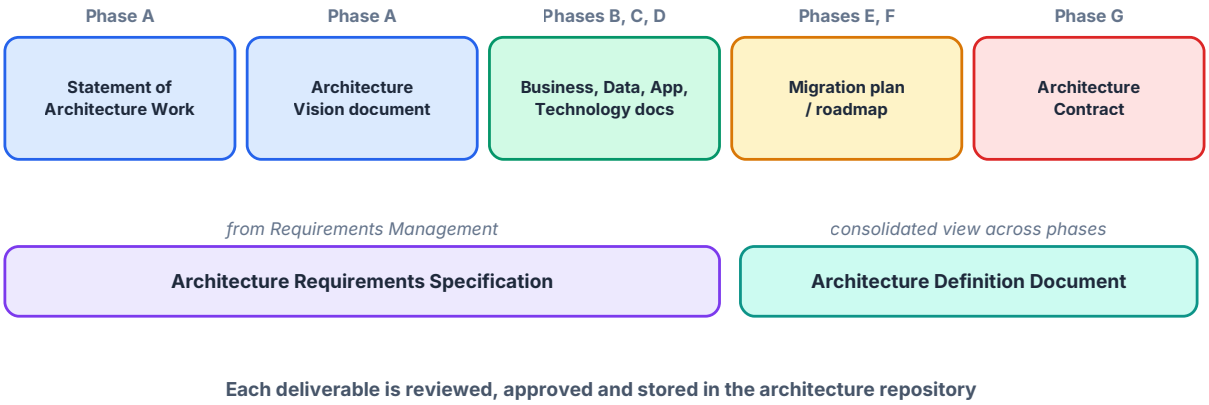


Fig. 57 — Common deliverables and the phases that produce them

Deliverable	What it is
Statement of Architecture Work (Phase A)	Formal agreement to begin the architecture project
Architecture Vision document (Phase A)	High-level, aspirational description of the target state
Business, Data, Application, Technology Architecture documents (B, C, D)	Detailed target-state descriptions per domain
Architecture Requirements Specification	Consolidated, validated requirements (from Requirements Management)
Migration Plan / Implementation Roadmap (E, F)	Sequenced plan from baseline to target architecture
Architecture Contract (Phase G)	Formal agreement between architecture and implementation teams that delivered systems conform
Architecture Definition Document	Comprehensive consolidated view of the overall target architecture across phases

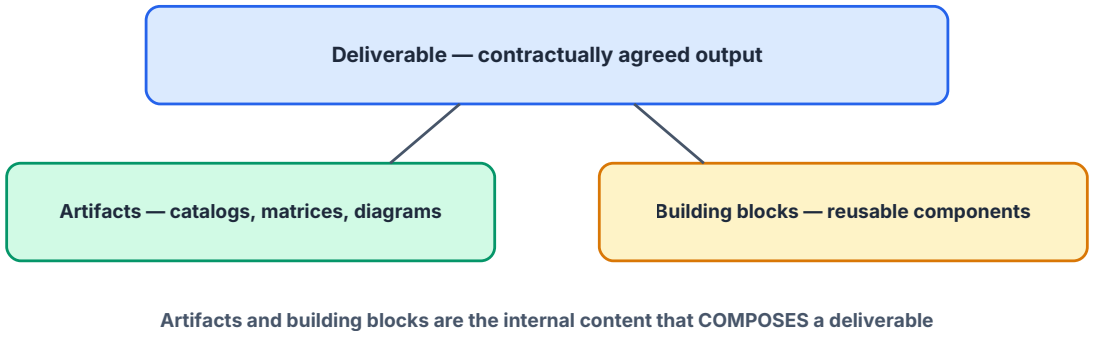


Fig. 58 — Deliverables are composed of artefacts and building blocks

Deliverables vs artefacts vs building blocks: a deliverable is typically a contractually agreed output; artefacts (catalogs, matrices, diagrams) and building blocks (reusable components, Sec. 5.9) are the internal content that composes it.

🧠 **Yaad rakho**

Deliverable = **agreed output**. Artifact = **chhota kaam (catalog, matrix, diagram)**. Building block = **dobara use hone wala tukda**.

5.7 Enterprise Continuum & Architecture Repository

📘 **Definition**

The **Enterprise Continuum** is a conceptual tool that classifies all architecture and solution assets on a spectrum from **generic to organisation-specific**.

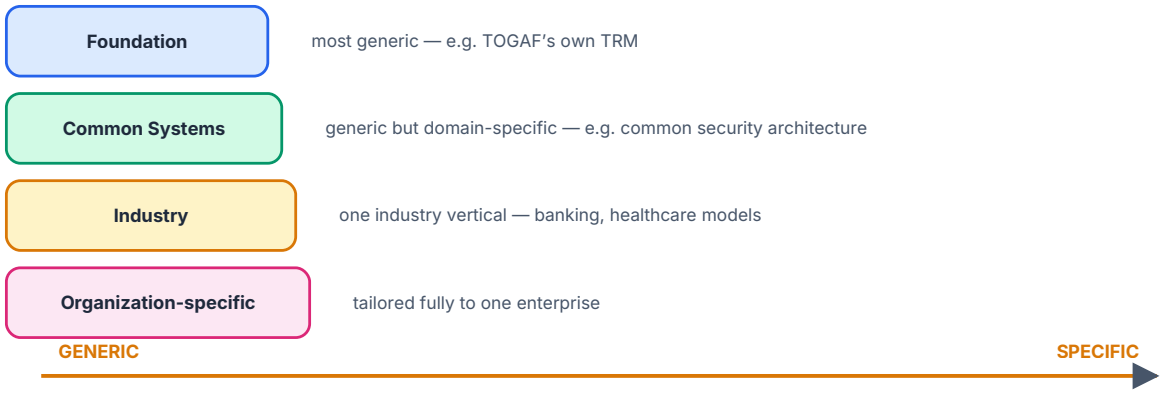


Fig. 59 — The four levels of the Enterprise Continuum, from generic to specific

Level	Meaning
Foundation Architectures	Most generic, broadly applicable — e.g. TOGAF's own Technical Reference Model
Common Systems Architectures	Generic but for a specific domain — e.g. "common security architecture"
Industry Architectures	Specific to an industry — e.g. banking or healthcare reference models
Organization-Specific Architectures	Most specific — tailored to one enterprise's needs

The idea: instead of building every architecture from scratch, architects **reuse and specialise** assets as they move along the continuum — saving time and increasing consistency.

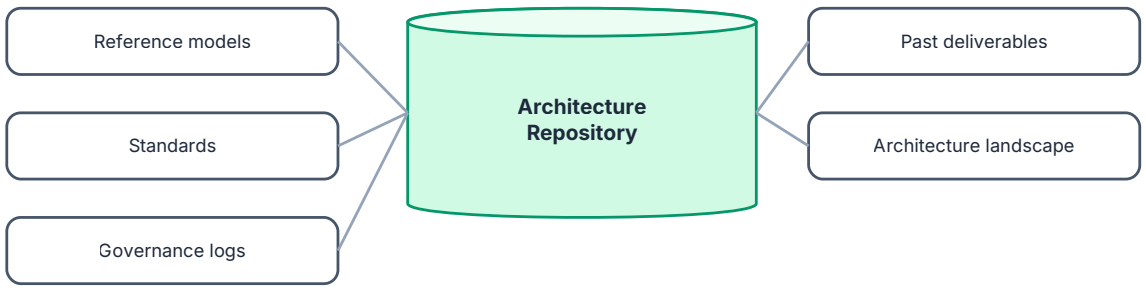


Fig. 60 — The Architecture Repository stores outputs, organised by the Continuum

The **Architecture Repository** is the actual storage that holds all architecture outputs — reference models, standards, governance logs, past deliverables and the organisation's architecture landscape — organised by the Enterprise Continuum's classification.

🧠 **Yaad rakho**

Continuum = **generic se specific tak ka scale** (Foundation → Common Systems → Industry → Organization). Repository = **godown** jahan sab rakha jaata hai.

TOGAF as an Enterprise Architecture Framework — official, The Open Group

5.8 Architecture Governance

📘 **Definition**

Architecture Governance ensures architecture work is executed consistently, complies with established principles and standards and stays aligned with business objectives — throughout the whole ADM lifecycle, not just at the end.

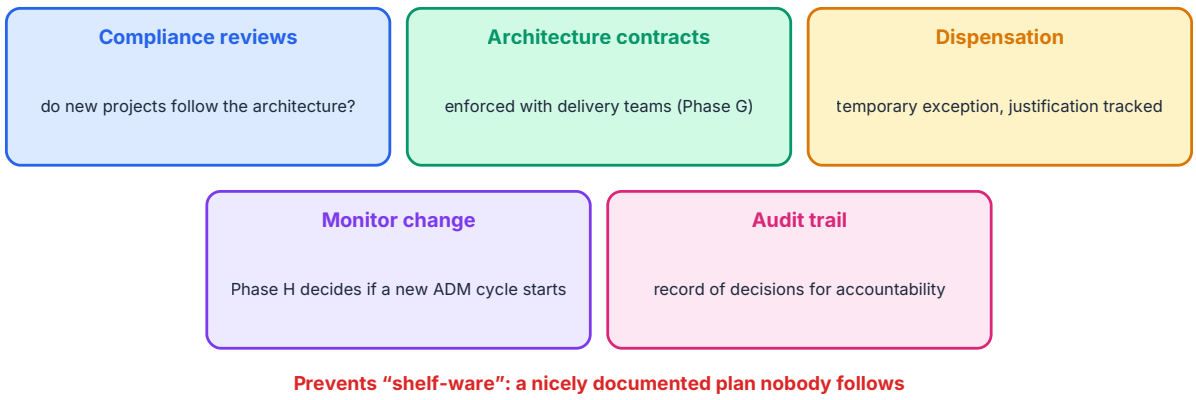


Fig. 61 — Core governance activities

- **Compliance reviews** — new systems/projects conform to the approved architecture.
- **Managing architecture contracts** — enforcing agreements with delivery teams (Phase G).
- **Dispensation process** — formal temporary exceptions with tracked justification.
- **Monitoring change** — via Phase H, deciding when changes trigger a new ADM cycle.
- **Audit trail** of architecture decisions for accountability.

> 💡 **Why it matters**
>
> Good governance stops an enterprise architecture from becoming **"shelf-ware"** — a nicely documented plan nobody follows.

5.9 Views, Viewpoints & Architecture Building Blocks

> 📘 **Definition**
>
> A **viewpoint** is a specification/template defining how to construct and use a particular kind of view — the lens. A **view** is the actual representation created by applying a viewpoint to real architecture content.

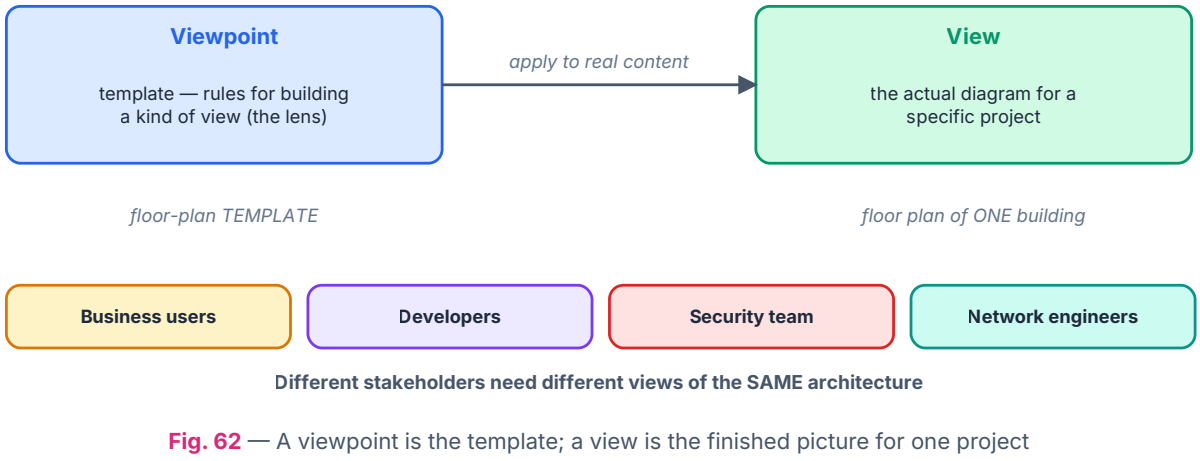


Fig. 62 — A viewpoint is the template; a view is the finished picture for one project

Analogy: a viewpoint is the blueprint *template* for how to draw a floor plan; a view is the actual floor plan of a specific building. Different stakeholders — business users, developers, security teams, network engineers — need different views of the same architecture.

Architecture Building Blocks

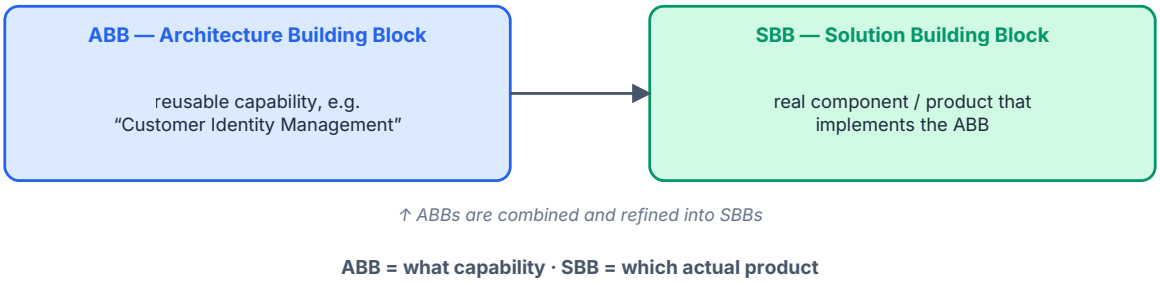


Fig. 63 — ABBs are refined into SBBs

Architecture Building Blocks (ABBs) are reusable, well-defined components of architecture — potentially reusable capabilities such as "Customer Identity Management" — that are combined and refined into **Solution Building Blocks (SBBs)**, the real-world components/products implementing the ABB in a specific solution.

🧠 **Yaad rakho**

Viewpoint = **template**, View = **asli diagram**. ABB = **kya capability chahiye**, SBB = **kaun sa product laga**.

❓ **Likely exam questions**

- What is TOGAF? State its key facts and BDAT levels.
- Explain the management benefits of TOGAF 9.
- List and explain the components of TOGAF 9.
- What is the ADM? Draw and explain the ADM cycle.
- Explain the ADM phases with their purpose.
- What are ADM deliverables? Differentiate deliverables, artefacts and building blocks.
- Explain the Enterprise Continuum and the Architecture Repository.
- What is architecture governance? Explain dispensation and compliance reviews.
- Differentiate view and viewpoint. What are ABBs and SBBs?

The TOGAF Standard 9.2 — official documentation, The Open Group · opengroup.org/togaf

Quick comparison sheet

One-glance tables for the last day before the exam. Each row is a favourite "differentiate between" question.

Units I–II

Topic	Remember
UI / UX	Look and feel (visual layer) / whole journey — UX is broader
UX process	Research → Define → Ideate → Design → Test → Implement & Iterate
Design Thinking	Empathize → Define → Ideate → Prototype → Test
Design principles	Simplicity, user-centred, visibility, consistency, feedback, clarity, accessibility, efficiency, hierarchy
IA questions	Where am I? What's here? Where can I go next?
IA / UX	How content is structured / how the product feels overall
8 IA principles	Objects, Choices, Disclosure, Exemplars, Front Doors, Multiple Classification, Focused Navigation, Growth
Four systems of IA	Organization, Labeling, Navigation, Search
System / Application / Enterprise / Internet	One system / one app's internals / whole organisation / networked web
Open / closed card sort	Users create categories (discover) / sort into given ones (validate)

Units III–V

Topic	Remember
Exact / ambiguous schemes	Alphabetical, chronological, geographical / topical, task, audience, metaphor
Website / intranet	Broad external audience, simple schemes / known internal audience, department-based
Top-down / bottom-up	From business goals / from the content that exists
Navigation types	Hierarchical, Global, Local, Contextual
Integrated / remote	Inside the page template / sitemap, index, guided tour
Miller's rule	7 ± 2 options
Label roles	Contextual links, navigation, headings, index terms, icons
Double Challenge	Fit content and structure + make sense to real users
Search patterns	Known-item, existence, exploratory, comprehensive
Search engine parts	Web crawler, database/index, search interface
TOGAF	EA framework, The Open Group, 1995, BDAT
ADM	Preliminary, A–H around central Requirements Management
Enterprise Continuum	Foundation → Common Systems → Industry → Organization-specific
View / viewpoint	Actual diagram / template to build it
ABB / SBB	Reusable capability / real product implementing it

Exam-day revision checklist

Tick these off — if you can **write or draw** each one from memory, you are well covered.

- [ ] UI vs UX with the house analogy
- [ ] Eight design issues in UX/UI
- [ ] UX/UI process stages and tools; Design Thinking
- [ ] Design principles and the 12 visual-design principles
- [ ] IA definition, three questions, IA vs UX
- [ ] The 8 IA principles with one line each
- [ ] Role of the information architect
- [ ] Six areas of IA and the four systems
- [ ] System vs Enterprise vs Application vs Internet architecture
- [ ] Open vs closed card sorting; tree testing
- [ ] Exact vs ambiguous organization schemes
- [ ] Four organizational challenges
- [ ] Website vs intranet; top-down vs bottom-up
- [ ] Cohesive organization system — five practices
- [ ] Browser navigation features
- [ ] Building context vs improving flexibility
- [ ] Four navigation types (H-G-L-C)
- [ ] Integrated vs remote navigation elements
- [ ] Seven guidelines for elegant navigation; Miller's 7±2
- [ ] Five label roles; style rules; five label sources
- [ ] Fine-tuning labels; non-representational labels
- [ ] The Double Challenge
- [ ] Four search behaviours
- [ ] Eight search-interface guidelines; three search-engine components
- [ ] Reference interview; indexing the right stuff
- [ ] To search or not to search — rule of thumb
- [ ] TOGAF key facts, BDAT, management benefits
- [ ] Six TOGAF components
- [ ] ADM cycle and every phase's purpose
- [ ] ADM deliverables; deliverable vs artifact vs building block
- [ ] Enterprise Continuum levels and the Architecture Repository
- [ ] Architecture governance activities
- [ ] View vs viewpoint; ABB vs SBB

Recommended resources

Resource	Covers
Rosenfeld & Morville — *Information Architecture for the World Wide Web* (O'Reilly)	Units II–IV (the standard IA textbook)
Information Architecture: A Practical Guide	IA principles, labeling
Loop11 · Big Sea · CleverX articles	IA basics, organization systems, card sorting
The Open Group — The TOGAF Standard 9.2 (opengroup.org/togaf)	Unit V (ADM, deliverables, continuum, governance)
KnowledgeHut — 10 Phases of TOGAF 9.2 ADM · CIO.com — What is TOGAF?	Unit V overview
geeksforgeeks.org — UI/UX and IA articles	Units I–II, with extra practice

All diagrams in this book were drawn for this book. Per-topic "GFG" links open GeeksforGeeks **search** pages for extra reading. Cross-check university-specific lecture slides for local terminology variations.